Nonintrusive Cross-Motor Detection and Severity Assessment of False-Data-Injection Attacks in Heterogeneous Inverter-Fed Drives

Abolfazl Najar, He Yang, Jin Ye, *Senior Member, IEEE*, Ping Ma and WenZhan Song, *Senior Member, IEEE*

Comparison: main.tex → main_revised.tex

Blue underlined text is added or revised.
~~Red struck-out text is removed.~~

Unmarked text is unchanged. Tables and figures are paired as complete ORIGINAL and REVISED blocks (left/right or top/bottom). Cross-references follow the revised numbering. Formatting-only changes and commented-out source text are not marked word by word.

***Abstract*—**

**This paper presents ~~an experimentally validated method for detecting~~ a nonintrusive monitoring framework for false-data-injection attacks (~~FDIA) and estimating attack severity in heterogeneous motor-drive systems using measurements acquired either at the motor terminals or at the point-of-common-coupling. The method extracts multi-scale features from variable-size windows and uses an InceptionTime model for binary attack detection and four-level severity classification. Experiments on a two-branch platform containing one~~ FDIAs) in heterogeneous inverter-fed motor drives. The framework combines low-frequency, high-frequency, and time-domain electrical features with an InceptionTime classifier to detect attacks and classify the injected current-feedback bias. Motor-terminal currents and point-of-common-coupling (PCC) voltage and current are evaluated as separate sensing configurations. Experimental data are obtained from an alternating current induction motor (ACIM) ~~drive and one~~ and a permanent magnet synchronous motor (PMSM) ~~drive show that motor-side sensing yields the strongest class separability, while point-of-common-coupling (PCC) sensingremains attractive for nonintrusive deployment. In the mixed-domain setting, binary detection reaches 98.89%accuracy with motor-side data and 89.44% accuracy with PCC data. For four-class severity assessment, the motor-side stream achieves 98.68% accuracy, while the PCC stream achieves 84.17% accuracy. Among the evaluated deep-learning and classical machine-learning baselines, the proposed method achieved the best PCC-side performance for both binary detection and four-level severity assessment. For the PCC-side tasks, the proposed method outperformed the evaluated baselines, achieving 89.44% accuracy for binary detection and 84.17% accuracy for four-level severity classification , compared with the best baseline accuracies of 86.45~~under three speed and three load settings. Eighteen recordings are evaluated using three training–validation–test partitions with no shared recordings or attack episodes within a partition. Mean detection accuracy is 87.41% for motor-terminal sensing and 80.78% for PCC sensing; four-class severity balanced accuracy is 90.81% and ~~83.42%, respectively. In cross-motor experiments, few-shot fine-tuning improved transferperformance, especially in the~~ 55.62%, respectively. The PCC severity classifier detects 61.39%, 85.08%, and 88.09% of low-, medium-, and high-bias attacks, with a 37.23% false-alarm rate. These detection rates include attacks assigned to an incorrect severity class. Supervised PMSM-to-ACIM ~~direction. These results indicate that PCC-based FDIA monitoring is promising on the tested ACIM/PMSM platform.~~

adaptation achieves 91.38% PCC balanced accuracy using 20% of the target episode groups; ACIM-to-PMSM transfer remains less effective. For PCC detection, omitting high-frequency features reduces processing time from 23.03 to 12.69 ms with similar balanced accuracy. The results establish the sensing and computational tradeoffs of the proposed framework and identify the remaining limitations of PCC severity classification and cross-motor transfer.

***Index Terms*—False-data-injection attack, point of common coupling, inverter-fed motor drives, heterogeneous motor drives, nonintrusive monitoring, cross-motor adaptation, severity assessment, cyber-physical security**

Abolfazl Najar, He Yang, Jin Ye, and WenZhan Song are with the College of Engineering, University of Georgia, Athens, GA 30602, USA (e-mail: najar@uga.edu; heyang95@uga.edu; jin.ye@uga.edu; wsong@uga.edu).

Ping Ma ~~are~~is with the Department of Statistics, University of Georgia, Athens, GA 30602, USA (e-mail: pingma@uga.edu).

Corresponding author: Jin Ye (e-mail: jin.ye@uga.edu).

I. Introduction

FDIA detection and severity assessment are important in networked motor-drive systems, where manipulated sensor feedback can degrade control performance and cause unsafe operation [1]. Increasing drive connectivity makes timely detection of sensor-oriented cyber attacks essential for resilient operation [2], [3]. Current signals are attractive for FDIA monitoring because they are easy to acquire, strongly coupled to electromechanical dynamics, and have been shown to reveal sensor-integrity attacks in motor drives [4], [5]. However, most existing studies emphasize terminal-side sensing, operate on a single machine domain, or assume sufficient labeled data for supervised training. These assumptions limit practicality in heterogeneous deployments where different motor types, sensing locations, and data availability constraints all affect performance. Motivated by these limitations, this work experimentally evaluates FDIA detection in a heterogeneous motor-drive setting involving both ACIM and PMSM domains. The study evaluates binary detection ~~, four-level severity assessment , cross-domain transfer, and deployment feasibility~~(`fdia2`), four-class injected-bias assessment (`severity4`), cross-motor adaptation, and processing time for converter-fed industrial loads connected at a shared electrical interface. The framework combines low-frequency (LF), high-frequency (HF), and time-domain (TD) ~~current~~ electrical features with an InceptionTime classifier~~, and compares~~. Separate models are trained for motor-terminal and

PCC sensing. An auxiliary motor-type classifier (`motor2`) distinguishes ACIM and PMSM recording conditions.

A. Challenges at PCC in Heterogeneous Multi-Motor Systems

From a deployment perspective, motor-terminal measurements provide cleaner machine-specific signatures, whereas PCC measurements are more scalable and non-intrusive. PCC sensing is attractive because it avoids direct motor-terminal instrumentation and is compatible with shared electrical infrastructure [6]. However, PCC currents are affected by converter harmonics, network interactions, and load aggregation. Consequently, PCC attack signatures are weaker and less separable than motor-terminal signatures [7]. The problem becomes more difficult in heterogeneous multi-motor settings. ACIM and PMSM systems exhibit different current characteristics even under the same attack framework, which introduces a substantial domain shift between source and target deployments. Consequently, a detector trained on one motor family may generalize poorly to the other unless explicit adaptation is used. Therefore, this study evaluates in-domain performance, source-only transfer, few-shot adaptation, and transfer-direction asymmetry.

B. Contributions

The main contributions of this work are as follows:

- **Developed** and validated a A nonintrusive PCC-based FDIA monitoring framework for heterogeneous inverter-fed drives, showing that PCC sensing supports useful binary detection and coarse four-level severity assessment without direct motor-terminal instrumentation.
- **Constructed** a multi-scale LF/HFis experimentally evaluated on a heterogeneous ACIM/TD current-feature representation and showed that it improves PCC-side separability under harmonic distortion and signal aggregation, while also enabling stronger motor-side performancePMSM drive platform. Motor-terminal and PCC sensing are assessed using separate models and complete recordings reserved for testing.
- **Provided** a cross-motor evaluation across ACIM and PMSM domains, including source-only transfer Attack detection and injected-bias classification are evaluated separately to characterize sensitivity to low-, few-shot target-domain adaptation, and latency/pruning analysis, showing both the limits of direct transfer and the practical gains from lightweight adaptationmedium-, and high-bias attacks. Classwise detection rates, severity-classification recall, and false-alarm rates are reported over three recording partitions.
- Cross-motor transfer and supervised adaptation are evaluated in both directions. Feature-family ablation and validation-based feature selection quantify the tradeoff between detection performance and processing time.

The remainder of this paper is organized as follows: Section II reviews related literature. Section III describes the physical system and dataset construction, while Section IV details the proposed FDIA detection framework and feature pipeline. Experimental results and cross-domain adaptations are reported in Section V. Section VI analyzes real-time deployment and latency, followed by a discussion of tradeoffs in Section VII. Section VIII concludes the paper.

II. Related Work

FDIA detection has been studied extensively in smart grids and broader cyber-physical systems (CPS), but the assumptions in that literature do not transfer cleanly to their system models and measurement requirements differ from those of inverter-fed motor drives. Many grid- and CPS-oriented approaches rely on state estimation, observer-based detection, invariant analysis, adaptive neuro-fuzzy models, or graph-based learning [8], [9], [10], [11], [12], [13]. These studies demonstrate strong FDIA-detection capability at the system level, but they are generally not designed for drive-level current monitoring or for deployment at a shared point of common coupling (PCC).

Motor-drive-specific studies are more closely related to the present work. Prior work has shown that current signals can support effective FDIA detection in electric-drive systems and that transfer learning can improve performance when labeled data are limited [2], [3]. Related industrial-IoT monitoring studies have also considered cyber-detection in machine-monitoring settings [14]. However, these studies mainly rely on motor-terminal sensing, single-motor settings, or closely related operating domains, and therefore provide limited evidence for heterogeneous cross-motor transfer.

The main gap is not FDIA detection in general, which has been widely studied in smart grids and cyber-physical systems, but nonintrusive PCC-based monitoring for heterogeneous inverter-fed motor-drive systems. In particular, the existing literature does not clearly establish: 1) how PCC sensing compares with motor-side sensing for low-severity current-feedback FDIAs, 2) whether PCC measurements support reliable multi-level severity assessment rather than only binary detection, given that attack signatures become weaker and less separable under signal aggregation and harmonic distortion, and 3) how detection performance transfers between heterogeneous ACIM and PMSM domains when labeled target-domain data are limited. The emphasis of the present work is the experimental assessment of sensing location, attack magnitude, and cross-motor adaptation using a common classification framework. This work addresses these three questions experimentally on a shared ACIM/PMSM platform using multi-scale features, in-domain evaluation, source-only cross-domain transfer, and few-shot target-domain adaptation.

III. System and Data

A. Experimental Platform: ACIM/PMSM and PCC Sensing

Fig. 1 illustrates the overall setup, including the shared PCC, the two drive branches, the ACIM and PMSM loads, the sensing locations, and the FDIA injection points in the control/feedback paths. The experimental platform consists of a heterogeneous two-motor industrial setup including one ACIM branch and one PMSM branch. Both branches are supplied

from a common upstream electrical interface, enabling measurements at the PCC in addition to motor-side measurements. This setup enables comparison of motor-terminal and PCC sensing under heterogeneous ACIM/PMSM dynamics. ~~A total of~~ The dataset contains 540 ~~unique FDIA cases were collected, including 270 ACIM cases and~~ scheduled attack episodes (270 ~~PMSM cases~~for each motor), formed by repeating 30 bias patterns over nine speed/load combinations per motor. All episodes were acquired on the same two-motor platform. The experiments cover three speed operating points, namely 900, 1100, and 1300 ~~rpm~~r/min.

Two measurement streams were used throughout this study. The first stream corresponds to motor-side measurements, referred to as *motor* in the remainder of the paper. The second stream corresponds to PCC-side measurements, referred to as *PCC*. The PCC stream is the primary deployment-oriented sensing location because it avoids dedicated sensors at each machine. The measured signals and channel counts associated with the motor-side and PCC-side streams are summarized in Table II.

B. Attack Model, Sensing Assumptions, and Labels

In the testbed, the FDIA is injected into the controller's phase-current feedback path. The PCC and motor-side monitoring sensors are used as trusted measurements to record the physical response of the drive system under attack. The monitoring sensors are isolated from the compromised controller feedback path. Thus, PCC sensing represents the nonintrusive monitoring case, while motor-side sensing serves as a performance reference. Replay, denial-of-service, monitoring-sensor attacks, and malicious command manipulation are outside the scope of this study.

Let $\mathbf{i}_{abc}(t) = [i_a(t), i_b(t), i_c(t)]^\top$ denote the true three-phase current-feedback vector in the controller's per-unit representation, and let $\tilde{\mathbf{i}}_{abc}(t)$ denote the corrupted current-feedback vector received by the motor controller. During an attack interval, the FDIA is modeled as an additive phase-current bias:

$$\tilde{\mathbf{i}}_{abc}(t) = \mathbf{i}_{abc}(t) + \boldsymbol{\alpha}, \qquad t \in [t_{\text{on}}, t_{\text{off}}], \tag{1}$$

Here, $\boldsymbol{\alpha} = [\alpha_a, \alpha_b, \alpha_c]^\top$ denotes the injected per-unit bias vector, with $\boldsymbol{\alpha} = \mathbf{0}$ outside the attack interval. The tested nonzero bias magnitudes range from 0.0005 to 0.0050 pu, corresponding to 0.05%–0.5% current-feedback bias. ~~These low-severity FDIAs are difficult to detect because their signatures can be masked by normal variation, harmonics, and PCC aggregation~~The bias is normalized to the nominal controller current-feedback base. In this study, severity is defined by the injected bias magnitude; physical consequences such as shaft-torque deviation are not used as severity labels. Attack severity is measured by the largest absolute injected phase-current bias, $s = \|\boldsymbol{\alpha}\|_\infty$.

For binary FDIA detection, each window is labeled as $y_{\text{fdia2}} = 1$ if $s > 0$ and $y_{\text{fdia2}} = 0$ otherwise. For four-class severity assessment, the label is defined as

$$y_{\text{severity4}} = \begin{cases} 0, & s = 0 \quad \text{(normal)}, \\ 1, & 0.0005 \le s \le 0.0015 \quad \text{(mild)}, \\ 2, & 0.0020 \le s \le 0.0035 \quad \text{(medium)}, \\ 3, & 0.0040 \le s \le 0.0050 \quad \text{(severe)}. \end{cases}$$

$$y_{\text{severity4}} = \begin{cases} 0, & s = 0 \quad \text{(normal)}, \\ 1, & 0.0005 \le s \le 0.0015 \quad \text{(low)}, \\ 2, & 0.0020 \le s \le 0.0035 \quad \text{(medium)}, \\ 3, & 0.0040 \le s \le 0.0050 \quad \text{(high)}. \end{cases} \tag{2}$$

~~corresponding to normal,mild,~~The class boundaries partition the tested discrete bias values and do not represent equipment protection limits. Cases 21–30 have $s = 0.0050$ throughout because phase A remains fixed; all belong to the high-bias class. The resulting attack-window counts are 5,400 low, 7,200 medium, and ~~severe conditions,respectively~~14,400 high. Table I summarizes the swept single-phase, equal two-phase, and fixed-plus-swept two-phase FDIA cases. Across three speeds, three load settings, and two motor types, the dataset contains 540 total attack cases. Each recording ~~consists of~~ contains 30 consecutive 20-s episodes, with 10 s normal operation followed by 10 s FDIA operation. After the scheduled 600 s, the injected bias was zero and the motor continued running. All 6,054 complete post-schedule windows are retained as normal, giving 33,054 normal and 27,000 attack windows in total.

C. Data Acquisition and Operating Conditions

Electrical measurements were acquired from the testbed using a National Instruments cDAQ-9185 chassis and a custom data-collector interface board. The interface board routed synchronized motor-side and PCC-side analog channels to the acquisition system. Three brake-excitation settings were tested for each machine: 1–3 V for the ACIM branch and 2–4 V for the PMSM branch. Because torque was not directly measured, these brake-excitation voltages are used as load-condition labels. All channels were sampled at 10 kHz and stored in TDMS format. For learning-based analysis, each record was segmented into non-overlapping 0.2 s windows containing 2,000 samples. ~~This non-overlapping segmentation prevents temporal leakage before train/validation~~Non-overlap prevents reuse of raw samples between adjacent windows, but does not remove within-recording dependence. Therefore, each entire recording, including its 30 episodes and post-schedule normal segment, is assigned to one partition. For each motor, five recordings are used for training, two for validation, and two for testing. Three seeds (1234, 2234, and 3234) change both the recording assignment and initialization. All feature normalization statistics are fitted using training recordings only. All windows from a given speed/~~test splitting~~load recording, including adjacent attack episodes and the post-schedule normal interval, therefore remain in the same subset. The holdouts assess new recorded operating combinations on the same apparatus; recordings can recur

across repeats. The resulting dataset dimensions, input configurations, severity-label distributions, and evaluation protocols are summarized in Table III.

IV. The Proposed Method

A. LF/HF Feature Extraction Pipeline

To represent both slow electromechanical behavior and faster attack-induced disturbances, a two-scale LF/HF feature pipeline was applied independently to each ~~current~~ measured channel. For every sample, a 2,000-point outer window was first extracted from the raw 10 kHz recording. Within this outer window, two parallel sliding analyses were performed. The LF branch uses a 500-sample window with a stride of 25 samples, while the HF branch uses a 250-sample window with a stride of 29 samples. The LF and HF settings both produce 61 time steps, enabling concatenation into one temporal feature tensor. Fig. 2 illustrates ~~representative attack-onset responses and the corresponding~~ the motor-current and PCC-current waveforms together with representative LF/HF windows. ~~In both cases, the raw signal is shown around the attack-transition instant, and representative LF and HF windows are overlaid to illustrate the two-scale extraction procedure. The motor-side signals exhibit stronger and more localized variations, whereas PCC-side responses are more distributed due to their aggregated nature~~Time is referenced to the onset of the injected current-feedback bias. The overlaid frames show the two temporal scales used for feature extraction, while the lower panels show how selected features vary along the signal. These examples illustrate the extraction procedure; detection and severity performance are evaluated separately on held-out recordings.

In total, 37 features were extracted per channel, including 18 LF spectral features, 11 HF spectral/envelope features, and 8 TD features. The LF branch targets slow electromechanical changes, fundamental-frequency behavior, harmonic distortion, sideband effects, and low-band energy redistribution. The HF branch captures higher-frequency redistribution and envelope-related changes that may appear during small current-feedback FDIAs. The TD features provide complementary sensitivity to amplitude variation, impulsiveness, waveform asymmetry, and short-term signal complexity. ~~Together, these features make weak PCC FDIA signatures more separable despite harmonics, grid interaction, and aggregation~~Their relative contributions are evaluated through fixed-family ablations; improved separability is assessed empirically rather than assumed for every feature family.

B. Inception-Based Detection Model

The extracted LF/HF tensors were modeled using a one-dimensional InceptionTime architecture [15] for multivariate time-series classification. The input dimension is $(61, 222)$ for the motor stream and $(61, 74)$ for the PCC stream. As illustrated in Fig. 3, the network uses bottleneck compression, parallel one-dimensional convolutional kernels of lengths 39, 19, and 9, residual shortcuts, global average pooling, and a dense output layer. ~~Motor and PCC models used stream-specific hyperparameters selected by~~ Motor-terminal and PCC measurements are represented by separate input tensors and processed by separately trained models. Hyperparameters are selected independently for each sensing configuration using validation balanced accuracy. The Inception-based design is well suited to the proposed LF/HF representation because the extracted features retain temporal ordering across the 61-step sequence. This preserves temporal interactions among LF, HF, and TD features.

C. Architecture of Online FDIA Detection and Attack-Severity Estimation

~~The proposed pipeline supports online monitoring from trusted motor-terminal or PCC current measurements.Figure 4 illustrates the overall architecture~~ Fig. 4 shows the acquisition, feature-extraction, and inference stages of the proposed monitoring ~~pipeline. In the deployed setting, fixed-length windows of current data are acquired from either the motor terminals~~ framework. Two independent sensing configurations are considered: six motor-terminal current channels, or the PCC voltage and current channels. The signals from these two configurations are not concatenated. For each configuration, a 2,~~passed through the~~ 000-sample measurement window is converted into a 61-step multivariate LF/HF~~feature-extraction stage, and then evaluated by the selected learning modelto produce anomaly or severity decisions. The measurement, feature-extraction, and inference stages are modular, allowing the same pipeline to operate with motor-side or PCC inputs and supporting feature pruning, reduced-input models , and cross-motor adaptation. The framework provides~~ /TD feature sequence, normalized using its training-set statistics, and applied to the corresponding trained model. Separate classification models provide binary FDIA decisions and severity classes; an auxiliary regression model estimates the injected bias. The monitoring interface reports alarms, severity indicators, and ~~logs for supervisory systems; automatic~~ recorded events. Automatic mitigation is outside the ~~scope. The same architecture is used for binary detection and severity classification. Binary detection is~~ experimental scope. Binary detection provides the primary PCC ~~deployment task; severity estimates provide secondary~~ monitoring output, while severity classification supplies additional diagnostic information. ~~The experimental results in later sections show that motor-terminal sensing provides cleaner discriminative signatures, whereas PCC sensing offers a less intrusive sensing option for the tested setup.~~

D. Training Details and Hyperparameters

~~For the InceptionTime-based models, hyperparameter selection was performed separately for the motor and PCC streams using validation-set balanced accuracy. The search varied depth , filters~~A fixed sample of 100 candidate configurations was evaluated independently for each task (binary detection or four-class severity), sensing stream, and recording split, giving 1,~~bottleneck size, dropout~~ 200 training runs for hyperparameter selection. Candidate ranges were depth 3/4/6, filters 16/24/32/48, bottleneck width 8/16/32, dropout 0.20/0.35/0.50, learning rate 5×10^{-4} or 10^{-3},

ORIGINAL — Figure 1 (main.tex)

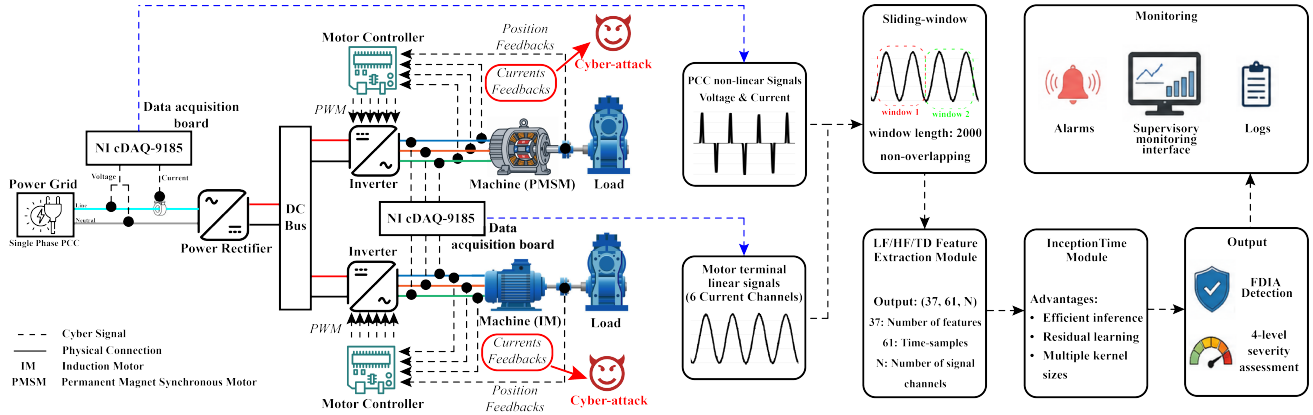


Figure 1: Experimental platform used to generate current-feedback FDIAs in a heterogeneous inverter-fed motor-drive testbed.

REVISED — Figure 1 (main_revised.tex)

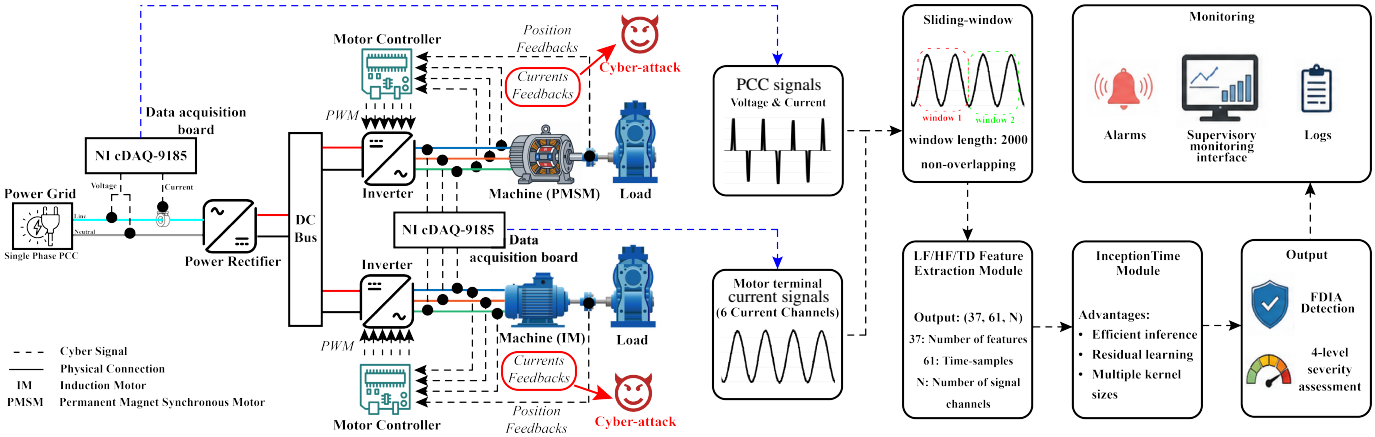


Fig. 1. Shared-rectifier, two-branch inverter-fed drive platform and monitoring locations. Motor-terminal currents and PCC voltage/current provide the monitoring inputs. The attack modifies controller-current feedback; the monitoring measurements are assumed trusted.

ORIGINAL — Table 1 (main.tex)

Table 1: FDIA dataset structure.

Cases	Phases	Bias Pattern	Total
1–10	A	α_a: 0.0005–0.0050	180
11–20	A,B	$\alpha_a = \alpha_b$: 0.0005–0.0050	180
21–30	A,B	$\alpha_a = 0.0050$, α_b: 0.0005–0.0050	180
		Overall	540

Each case group is evaluated over 3 speeds, 3 load settings, and 2 motor types. For all cases, unspecified phase biases are zero.

REVISED — Table 1 (main_revised.tex)

TABLE I
FDIA DATASET STRUCTURE.

Cases	Phases	Bias Pattern	Total
1–10	A	α_a: 0.0005–0.0050	180
11–20	A,B	$\alpha_a = \alpha_b$: 0.0005–0.0050	180
21–30	A,B	$\alpha_a = 0.0050$, α_b: 0.0005–0.0050	180
		Overall	540

Each case group is evaluated over 3 speeds, 3 load settings, and 2 motor types. For all cases, unspecified phase biases are zero.

ORIGINAL — Table 2 (main.tex)

Table 2: Measurement streams used for monitoring.

Stream	Measured signals	Channels
Motor-side	ACIM and PMSM phase currents	6
PCC-side	PCC voltage and current	2

REVISED — Table 2 (main_revised.tex)

TABLE II
MEASUREMENT STREAMS USED FOR MONITORING.

Stream	Measured signals	Channels
Motor-side	ACIM and PMSM phase currents	6
PCC-side	PCC voltage and current	2

ORIGINAL — Figure 2 (main.tex)

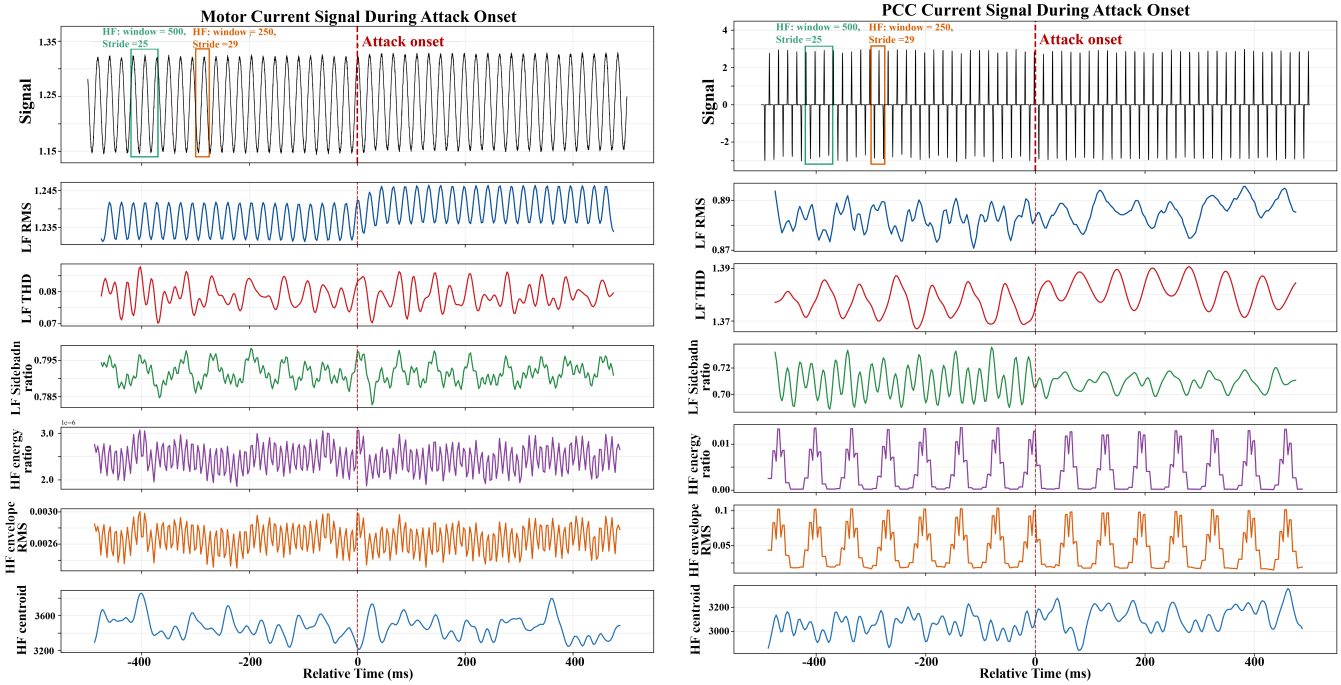


Figure 2: Representative motor-side and PCC-side signals at attack onset with the corresponding LF/HF feature responses.

REVISED — Figure 2 (main_revised.tex)

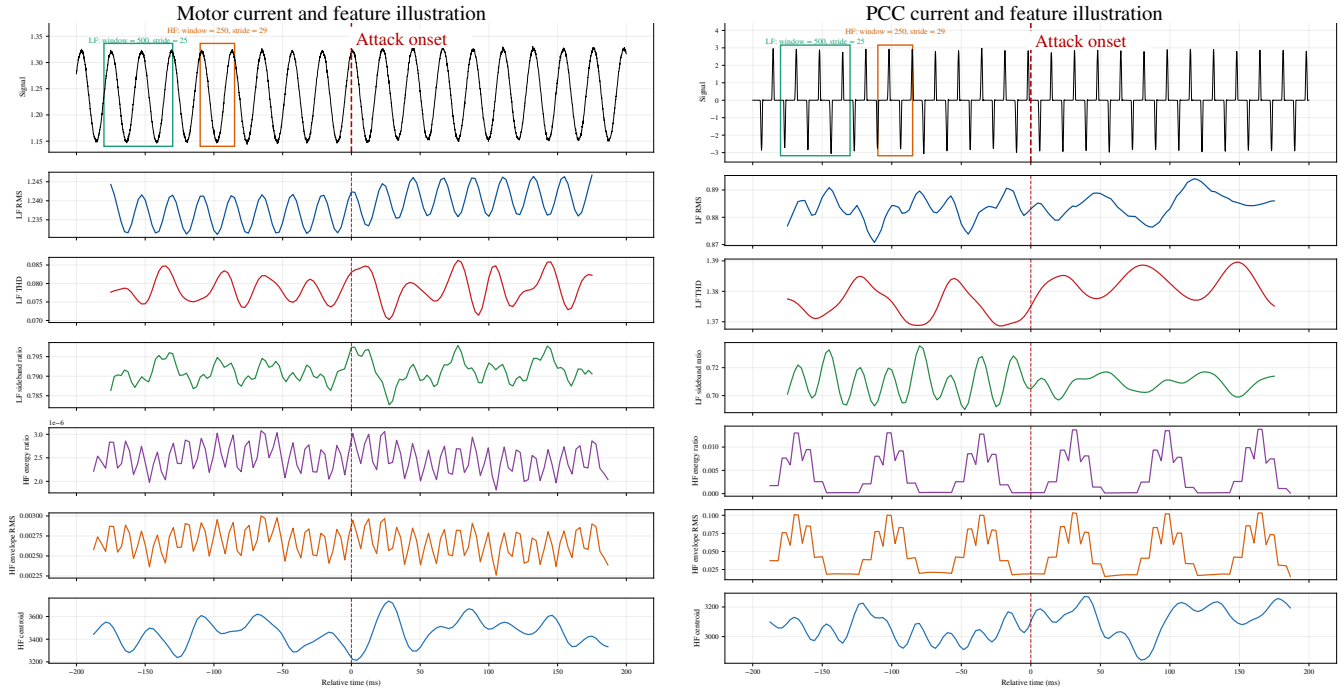


Fig. 2. Motor-current (left) and PCC-current (right) signal and feature illustrations. LF frames use 500 samples with stride 25; HF frames use 250 samples with stride 29. Zero denotes attack onset. The traces illustrate the electrical response and feature extraction around the onset; detector performance is evaluated separately on held-out recordings. Signal amplitude is in recorded acquisition units.

weight decay 5×10^{-5} or 10^{-4}, and batch size 64 or 128. The search evaluates the sampled configurations rather than the complete Cartesian product. Training used Adam, inverse-frequency weighted cross-entropy, a 60-epoch cap, and early stopping after 10 epochs without improvement in validation balanced accuracy. A plateau scheduler halved the learning rate after three unimproved epochs, down to 10^{-6}. Checkpoints and hyperparameters were selected only by validation balanced accuracy; ties followed the fixed candidate order. Table IV ~~summarizes the top-ranked configurations for both sensing streams~~illustrates the ranking for one specified split.

Reported test classification metrics are mean ± sample standard deviation over the three holdouts. Precision, recall, and F1 are macro-averaged; balanced accuracy is the mean class recall. Because the same recordings can recur across repeats, these standard deviations describe split/initialization sensitivity rather than uncertainty from independent physical replications. Test results were not used to select configurations or feature subsets. One-factor sensitivity experiments changed depth to 3/9, dropout to 0.1/0.4, or learning rate to $3\times10^{-4}/3\times10^{-3}$ around each selected configuration; all other settings were retained (Fig. 10).

E. Domain Adaptation Strategy: Source-Only vs Few-Shot Fine-Tuning

~~Few-shot adaptation was evaluated by training on one motor type and testing on the other. Two transfer directionswere considered:~~

Source-only transfer was evaluated in both directions, ACIM → PMSM and PMSM → ACIM. For each direction, ~~the detector was first trained using labeled data from the source motor onlyand then evaluated directly on the target motor. This *source-only* setting quantifies cross-motor domain shift.~~ six source recordings were used for training and three for validation. The same 100-candidate search was performed using source validation only, independently of mixed-domain model selection. Six target recordings formed the adaptation pool and the remaining three formed the target test set. Source validation recordings and target test recordings remain excluded from model fitting and adaptation, respectively.

~~For adaptation, the source-trained model was fine-tuned using a small labeled subset of target-domain samples before testing on held-out target data. The target adaptation ratios considered in this work were~~ Nominal adaptation budgets of 1%, 5%, 10%, and 20% ~~of the available target-domain adaptation pool. The domain-adaptation strategy was implemented as supervised fine-tuning rather than unsupervised feature alignment. Accordingly, all reported adaptation gains reflect the benefit of limited labeled target-domain data under a practical transfer-learning setting. To ensure reliable comparison, all cross-domain experiments were repeated over three random seeds, and the results were reported using both accuracy and balanced accuracy.~~ select 2, 9, 18, or 36 complete episode groups from the 180-group target pool; the first realized group fraction is 1.11%. For sampling only, each post-schedule normal tail is attached to its parent recording's final episode group while retaining its normal label and actual timestamp. Consequently, window fractions can differ from nominal group fractions. Fine-tuning updates the source model for 20 fixed epochs at 0.2 times its selected learning rate, with source normalization frozen. No target validation labels or target-test labels are used. All conditions use three seeds.

~~This protocol is particularly important in the present study because ACIM and PMSM signals exhibit substantial domain differences despite being collected under the same FDIA framework. This tests whether the LF/HF-Inception representation can be adapted with limited target-domain labels .~~

V. Experimental Results

A. In-Domain Baselines: Motor vs PCC

Table V ~~reports mixed-domain in-domain performance with ACIM and PMSM samples in all splits. The results show that motor-side sensing provides more discriminative attack signatures than PCC sensing for both (`fdia2`) and (`severity4`). In addition, (`severity4`) is substantially more challenging than binary attack detection, especially at PCC .~~ summarizes performance when both motor types are represented in the training, validation, and test sets, with no shared recordings between these sets. Motor-terminal sensing achieves mean balanced accuracies of 88.26% for FDIA detection and 90.81% for severity classification; the corresponding PCC values are 80.29% and 55.62%. Motor-terminal measurements provide stronger separation of the injected-bias classes, whereas PCC sensing avoids dedicated instrumentation at each motor. The motor-terminal detection accuracies are 88.83%, 97.77%, and 75.64% across the three test partitions, indicating substantial sensitivity to the recorded operating conditions.

~~Thus, PCC-side severity estimates should be treated as coarse diagnostics,~~ For the separate binary detectors, normal false-alarm rates are $21.15\pm21.77\%$ for motor sensing and $14.82\pm5.06\%$ for PCC sensing; missed-attack rates are $24.61\pm7.17\%$ for PCC and $2.32\pm1.79\%$ for motor sensing. The false-alarm and missed-attack rates therefore remain relevant constraints on practical monitoring performance.

For the four-class PCC model, exact low/medium/high recall is 38.33%/50.58%/70.79%, whereas detecting those windows as any attack yields 61.39%/85.08%/88.09%. The complementary severity-conditioned attack recall is defined as

$$R_{\text{attack}|k}=\frac{\sum_{j=1}^{3}C_{kj}}{\sum_{j=0}^{3}C_{kj}},\qquad k\in\{1,2,3\},\tag{3}$$

where C is the confusion matrix. Exact-severity recall is $C_{kk}/\sum_j C_{kj}$. The first metric measures detection irrespective of the predicted attack level, whereas ~~motor-side measurements support finer severity discrimination. Overall, PCC monitoring is more challenging than motor-side monitoring, and severity assessment is harder than binary detection. These baseline findings motivate the subsequent cross-domain analysis, where the additional challenge of~~

ORIGINAL — Table 3 (main.tex)

Table 3: Dataset dimensions and evaluation protocols.

Parameter	**Value / Configuration**
Data dimensions	
Total samples	60,054
Motor distribution	ACIM: 30,204; PMSM: 29,850
Binary distribution	Normal: 33,054; Attack: 27,000
Input configuration	
Time steps per sample, W	61
Features per channel	37
Channels	Motor: 6; PCC: 2
Motor feature shape	$(N, W, F) = (N, 61, 222)$
PCC feature shape	$(N, W, F) = (N, 61, 74)$
Severity labels	
Raw severity levels	$0\%, 0.05\%, 0.1\%, \ldots, 0.5\%$
Four-class labels	Normal: 33,054; Mild: 8,100; Medium: 10,800; Severe: 8,100
Protocol 1: In-domain evaluation	
Split ratio	Train / validation / test = 70% / 15% / 15%
Sample counts	42,037 / 9,008 / 9,009
Domain composition	ACIM and PMSM samples appear in all splits
Protocol 2: Cross-domain evaluation	
Transfer directions	ACIM $\rightarrow$ PMSM; PMSM $\rightarrow$ ACIM
Adaptation ratios	1%, 5%, 10%, and 20% labeled target data
Reporting	Three seeds; accuracy and balanced accuracy

REVISED — Table 3 (main_revised.tex)

TABLE III
DATASET DIMENSIONS AND RECORDING-DISJOINT EVALUATION PROTOCOLS.

Parameter	Value / configuration
Data dimensions	
Complete windows	60,054; 2,000 samples (0.2 s) each
Recordings / episodes	18 recordings; 540 scheduled episodes
Motor distribution	ACIM: 30,204; PMSM: 29,850
Normal / attack windows	33,054 normal; 27,000 attack
Normal tail windows	6,054; included with parent recording
Input configuration	
Features per channel	18 LF; 11 HF; 8 TD
Temporal feature steps	61 per outer window
Motor input	61×222; six current channels
PCC input	61×74; voltage and current
Severity labels (`severity4`)	
Normal / low	33,054 / 5,400 windows
Medium / high	7,200 / 14,400 windows
Protocol 1: mixed-motor holdouts	
Recordings per motor	5 training; 2 validation; 2 test
Repeat 1 (seed 1234)	33,718 training; 13,279 validation; 13,057 test windows
Repeat 2 (seed 2234)	33,275 training; 13,250 validation; 13,529 test windows
Repeat 3 (seed 3234)	33,560 training; 13,227 validation; 13,267 test windows
Protocol 2: cross-motor adaptation	
Source recordings	6 training; 3 validation
Target recordings	6 adaptation-pool; 3 held-out test
Target adaptation budgets	1/5/10/20%: 2/9/18/36 complete episode groups, respectively, from 180 groups

Each recording and all its episodes remain in one partition within a repeat.

ORIGINAL — Table 4 (main.tex)

Table 4: Top five InceptionTime configurations ranked by validation balanced accuracy on `severity4`.

Rank	D	F	BN	Dr	LR	WD	BS	test BA (%)
Motor stream								
1	3	32	16	0.20	0.0010	5×10^{-5}	128	98.25
2	3	32	8	0.20	0.0010	10^{-4}	64	98.08
3	3	16	16	0.35	0.0010	5×10^{-5}	128	98.02
4	3	32	8	0.20	0.0010	5×10^{-5}	128	97.99
5	3	32	8	0.35	0.0010	10^{-4}	64	97.96
PCC stream								
1	6	32	32	0.20	0.0010	10^{-4}	128	80.22
2	6	48	8	0.20	0.0005	5×10^{-5}	128	79.95
3	3	48	32	0.35	0.0005	5×10^{-5}	64	79.84
4	4	32	32	0.35	0.0005	10^{-4}	64	79.56
5	4	48	32	0.50	0.0010	10^{-4}	128	79.40

D: depth; F: filters; BN: bottleneck; Dr: dropout; LR: learning rate; WD: weight decay; BS: batch size; test BA: test balanced accuracy.

REVISED — Table 4 (main_revised.tex)

TABLE IV
TOP FIVE CONFIGURATIONS BY VALIDATION BALANCED ACCURACY ON `SEVERITY4`, SEED 1234.

Rank	D	F	BN	Dr	LR	WD	BS	Val. BA
MOTOR stream								
1	3	16	8	0.50	0.001	10^{-4}	64	**93.26**
2	3	24	8	0.20	0.0005	5×10^{-5}	64	92.51
3	6	24	16	0.35	0.001	5×10^{-5}	128	92.48
4	3	32	8	0.20	0.001	5×10^{-5}	128	92.37
5	4	16	8	0.50	0.001	5×10^{-5}	128	92.21
PCC stream								
1	6	32	32	0.50	0.001	10^{-4}	64	**52.25**
2	6	48	32	0.35	0.001	5×10^{-5}	64	51.94
3	3	16	32	0.50	0.001	5×10^{-5}	64	51.44
4	4	48	32	0.50	0.001	10^{-4}	128	51.26
5	6	24	32	0.35	0.0005	10^{-4}	128	51.14

D: depth; F: filters; BN: bottleneck; Dr: dropout; LR: learning rate; WD: weight decay; BS: batch size. Bold: highest validation BA within each stream. Test scores are not used for ranking.

motor-type shift is introducedexact-severity recall requires the correct bias class. The same PCC severity model has a 37.23% mean normal false-alarm rate and 71.40% mean accuracy after collapsing its predictions to normal/attack. It is distinct from the separately trained binary detector. Fig. 6 and Table XIII report these complementary quantities without replacing four-class performance.

B. Comparison with Deep and Classical ML Baselines

To benchmark the proposed LF/HF Inception-based model, representative deep-learning and classical ML baselines

ResNet, LSTM-FCN, Random Forest, XGBoost, and MultiRocket [16] were evaluated on the PCC-based `fdia2` and `severity4` tasks, which were the most challenging in-domain settings. The baselines include ResNet same splits and corrected targets. The LSTM-FCN implementation combines convolutional and recurrent branches without squeeze-and-excitation blocks. ResNet used 40 epochs and batch size 64; LSTM-FCN used 40 epochs, batch size 128, eight LSTM units, dropout 0.5, and convolutional widths 64/128/64. Trees used the flattened $61 \times F$ tensors: Random Forest compared 300/600 trees and unrestricted/20 depth with balanced-subsample weights; XGBoost compared depth 4 with learning rate 0.05 and depth 6 with learning rate 0.03, using 1,MLSTM-FCN, Random Forest, and XGBoost500 estimators and 0.8 row/column subsampling. MultiRocket used approximately 50,while the proposed model uses the final stream-specific InceptionTime configuration selected by validation-based hyperparameter search. 000 transformed features and ridge penalties 0.1/1/10/100. Available candidate settings were selected using validation balanced accuracy. Search budgets differ between model families, so this is a comparison under the stated settings rather than an equal-budget optimization study.

As shown in Table VI, the proposed model achieved the highest accuracy, balanced accuracy , and InceptionTime has the highest mean PCC balanced accuracy for both tasks, and the highest mean accuracy and macro F1 score for binary detection. For severity, XGBoost has higher raw accuracy (70.45% versus 61.00%), while LSTM-FCN has slightly higher macro F1 (53.23% versus 52.96%). InceptionTime's severity advantage is confined to mean balanced accuracy (55.62% versus at most 52.38% among the evaluated baselines on both PCC tasks. The performance gap is larger for `severity4`, confirming that PCC-based severity discrimination is more challenging than binary FDIA detection and benefits from the proposed temporal LF/HF representationalternatives). These differences reflect the unequal class populations. The reported means describe performance over three repeated partitions; statistical significance is not established.

C. Cross-Domain Transfer Without Adaptation

Source-only transfer was evaluated by training

Table VII reports binary detection when a model trained on one motor type and testing directly on is applied directly to the other. Two transfer directions were considered: ACIM → PMSM and PMSM → ACIM. Table VII shows that source-only fdia2 transferfalls near chance balanced accuracy for both sensingstreams. For `fdia2`, the source-only results in Table VII show a substantial degradation relative to the mixed-domain baselines. This confirms a substantial ACIM/PMSM domain shift in the tested transfer setting. Thus, reliable cross-motor FDIA detection in this setting benefits from target-domain adaptation ACIM-to-PMSM transfer, balanced accuracy is 50.60% with motor-terminal sensing and 51.58% with PCC sensing. The reverse direction yields 36.08% and 56.00%, respectively. These results indicate limited transferability of the learned decision boundaries. The below-chance motor-terminal result in the PMSM-to-ACIM direction indicates systematic classification errors. Balanced accuracy is reported alongside accuracy because the unequal normal and attack populations can otherwise obscure this behavior. The following subsection evaluates adaptation using labeled target episodes.

D. Domain Adaptation Results: 1/5/10/20%

Table VIII reports fdia2 few-shot adaptation using 1%, 5%, 10%, and `fdia2` balanced accuracy before and after target-domain fine-tuning. At 20%labeled target data. In all cases, adaptation improved performance relative to the source-only setting, although the magnitude of the gain was strongly direction dependent. For ACIM-to-PMSM transfer, adaptation gives only modest gains, reaching 57.49% balanced accuracy for motor sensing and 60.06% for PCC at 20% target data. For , motor fine-tuning reaches 75.99% for ACIM-to-PMSM and 92.63% for PMSM-to-ACIM transfer, adaptation is much stronger, reaching 86.10% balanced accuracy for motor sensing and 91.16% for PCC at 20%target data. Overall, the adaptation results show that few-shot target-domain . PCC fine-tuning can improve transfer performance, but the gains are strongly direction dependent and are much larger in the PMSM → ACIM direction. This asymmetry indicates that cross-motor FDIA adaptation cannot be treated as a symmetric transfer problem; in practice, the pretraining motor domain materially affects PCC deployment performance reaches 56.30% and 91.38%, respectively, preserving a pronounced direction asymmetry.

The improvement depends on the transfer direction. At the 1% budget, motor ACIM-to-PMSM performance remains near the source-only result, whereas PMSM-to-ACIM performance improves substantially. Increasing the target budget generally helps, but the PCC ACIM-to-PMSM sequence is not monotonic. The source-only model provides the reference for quantifying the improvement obtained by updating its parameters with labeled target episodes.

E. Error Analysis: Confusion Matrices and FP/FN Behavior

Fig. 5 compares the in-domain confusion matrices for `fdia2` and `severity4`. For `fdia2`, motor-side sensing yields a strongly diagonal confusion matrix with low false-positive and false-negative rates of 0.75% and 1.56%, respectively, whereas PCC sensing increases these rates to 6.55% and 15.46% because the attack is observed only through

ORIGINAL — Figure 3 (main.tex)

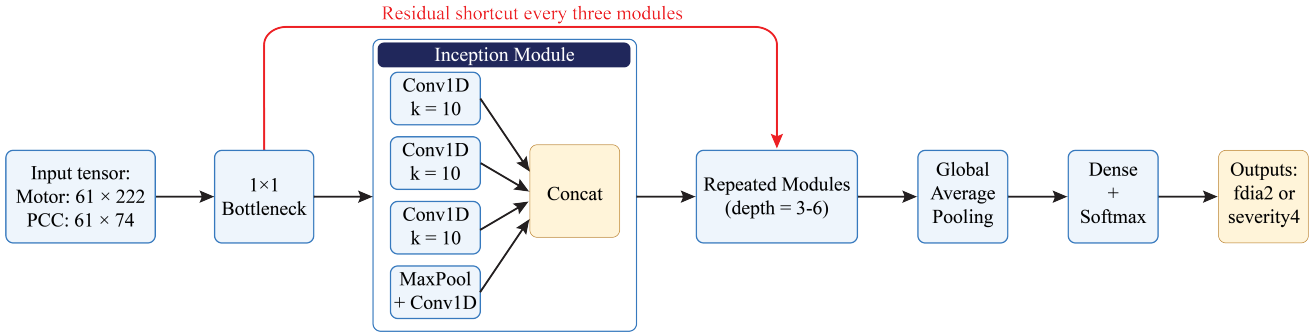


Figure 3: Overview of the InceptionTime-based classifier used for `fdia2` detection and `severity4` classification.

REVISED — Figure 3 (main_revised.tex)

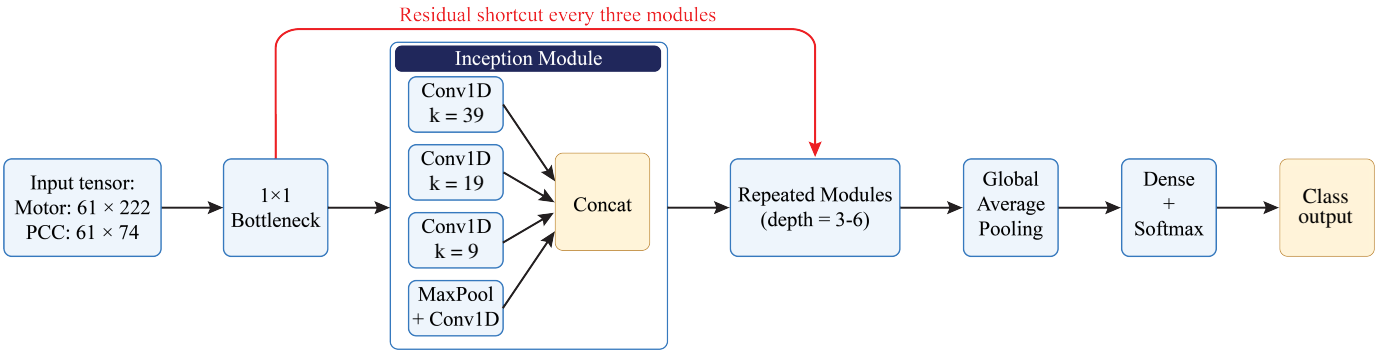


Fig. 3. InceptionTime classifier with a task-specific output: two classes for `fdia2` or `motor2`, and four classes for `severity4`. The convolution lengths are 39, 19, and 9. Each task and sensing configuration uses a separately trained model. Each module has three temporal convolutions and a max-pooling projection branch, followed by concatenation, batch normalization (BN), and ReLU. A projected residual shortcut is added after each group of three modules; depth and other settings are selected separately for each task, stream, and repeat.

ORIGINAL — Table 5 (main.tex)

Table 5: In-domain performance under the mixed ACIM/PMSM split.

Task	Accuracy (%)	Bal. Acc. Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
PCC stream					
`fdia2`	89.44	88.99	89.71	88.99	89.25
`severity4`	84.17	80.22	80.50	80.22	80.32
Motor type	100.00	100.00	100.00	100.00	100.00
Motor stream					
`fdia2`	98.89	98.85	98.91	98.85	98.88
`severity4`	98.68	98.25	98.22	98.25	98.24
Motor type	100.00	100.00	100.00	100.00	100.00

Precision, recall, and F1 are reported as macro-averaged values in paper.

REVISED — Table 5 (main_revised.tex)

TABLE V
MIXED ACIM/PMSM PERFORMANCE ON HELD-OUT RECORDINGS (%).

Task	Acc.	BA	Prec.	Recall	F1
PCC stream					
`fdia2`	80.78 (±2.95)	80.29 (±3.03)	81.04 (±3.14)	80.29 (±3.03)	80.42 (±2.96)
`severity4`	61.00 (±1.32)	55.62 (±2.94)	52.93 (±2.36)	55.62 (±2.94)	52.96 (±2.15)
`motor2`	94.40 (±9.70)	94.35 (±9.79)	95.35 (±8.06)	94.35 (±9.79)	94.27 (±9.92)
MOTOR stream					
`fdia2`	87.41 (±11.13)	88.26 (±9.99)	89.90 (±7.84)	88.26 (±9.99)	87.23 (±11.39)
`severity4`	87.64 (±8.87)	90.81 (±4.29)	86.09 (±9.44)	90.81 (±4.29)	86.06 (±8.95)
`motor2`	99.94 (±0.11)	99.94 (±0.11)	99.94 (±0.10)	99.94 (±0.11)	99.94 (±0.11)

Cells show mean above (± sample SD), over three holdouts; this is not a range or confidence interval. BA: balanced accuracy. Precision, recall, and F1 are macro-averaged.

ORIGINAL — Figure 4 (main.tex)

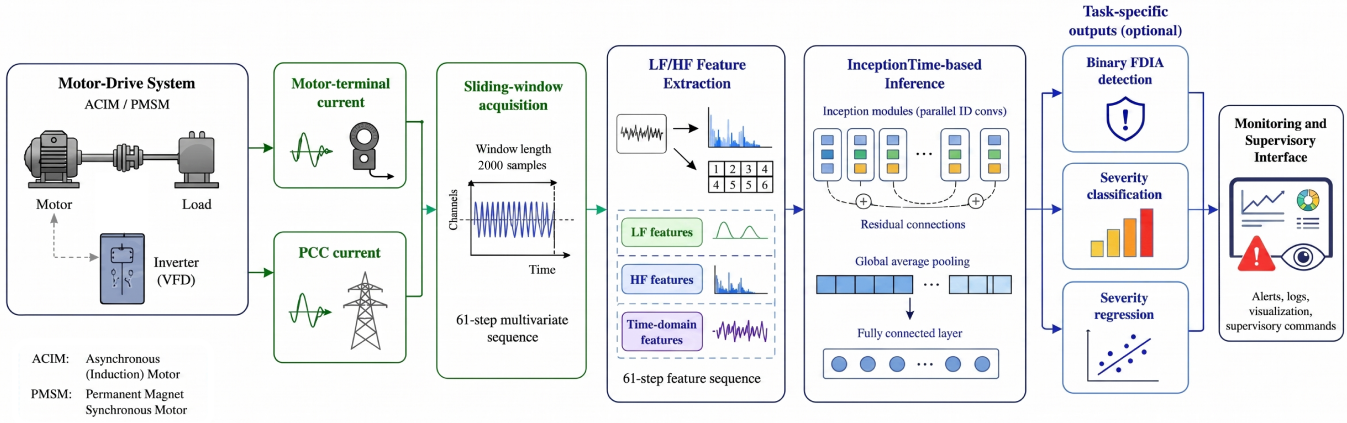


Figure 4: Online FDIA detection and severity-assessment architecture for motor-side and PCC-side sensing.

REVISED — Figure 4 (main_revised.tex)

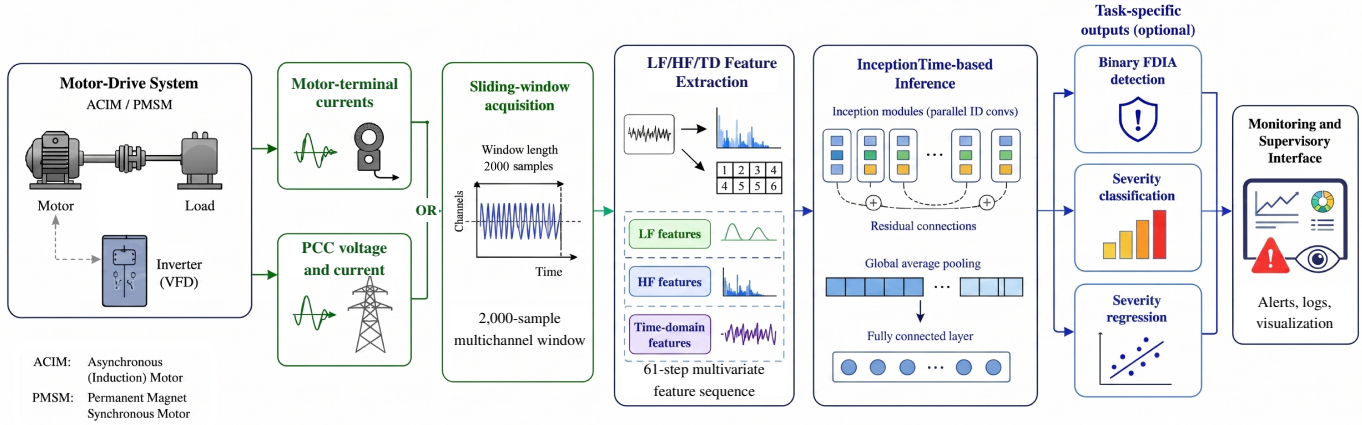


Fig. 4. Independent monitoring pipelines for motor-terminal and PCC sensing. The OR junction denotes alternative inputs: the six motor-current channels and the two PCC channels are processed separately, with separate normalization and trained models. Each 2,000-sample (0.2-s) window produces a 61-step multivariate feature sequence for task-specific inference. Regression is an auxiliary injected-bias estimate. The interface provides alerts, logs, and visualization; automatic mitigation is not evaluated.

ORIGINAL — Table 6 (main.tex)

Table 6: Baseline comparison on the PCC-based `fdia2` and `severity4` tasks.

Method	Acc. (%)	Balanced Acc. (%)	F1 (%)
fdia2			
Proposed InceptionTime	89.44	88.99	89.25
ResNet	85.58	84.83	85.20
MLSTM-FCN	84.58	83.80	84.17
Rand. Forest	86.05	85.27	85.67
XGBoost	86.45	85.75	86.11
severity4			
Proposed InceptionTime	84.17	80.22	80.32
ResNet	77.31	70.22	70.68
MLSTM-FCN	77.96	71.11	71.89
Rand. Forest	83.42	72.68	77.40
XGBoost	79.00	65.21	70.45

REVISED — Table 6 (main_revised.tex)

TABLE VI
PCC BASELINE COMPARISON ON `FDIA2` AND `SEVERITY4` (%).

Method	Accuracy	BA	Macro F1
fdia2			
InceptionTime	**80.78** ± 2.95	**80.29** ± 3.03	**80.42** ± 2.96
ResNet	73.91 ± 5.78	73.99 ± 5.33	73.76 ± 5.66
LSTM-FCN	72.57 ± 4.89	72.52 ± 4.31	72.35 ± 4.66
Rand. Forest	70.43 ± 1.00	68.69 ± 0.74	68.45 ± 0.85
XGBoost	75.37 ± 0.81	73.84 ± 1.11	73.97 ± 1.17
MultiRocket	70.10 ± 2.13	69.55 ± 2.82	69.35 ± 2.39
severity4			
InceptionTime	61.00 ± 1.32	**55.62** ± 2.94	52.96 ± 2.15
ResNet	65.18 ± 5.35	50.41 ± 5.44	51.15 ± 5.71
LSTM-FCN	66.60 ± 1.72	52.38 ± 2.12	**53.23** ± 1.94
Rand. Forest	67.36 ± 0.17	43.49 ± 2.18	45.07 ± 2.59
XGBoost	**70.45** ± 0.74	47.17 ± 2.02	49.67 ± 2.17
MultiRocket	58.41 ± 3.34	44.92 ± 3.46	44.28 ± 2.52

Mean ± sample SD over three holdouts. Bold: highest mean for each metric within a task; it does not indicate statistical significance.

the aggregated current response. For severity4, PCC errors concentrate between adjacent classes, especially shows the classwise errors for FDIA detection and severity classification. The motor-terminal detector has high attack recall, although its false-alarm rate varies substantially across recording partitions. The PCC detector has a higher missed-attack rate. For PCC severity classification, errors include both normal/mild and mild/medium, indicating that PCC measurements retain coarse attack information but blur finer severity distinctionsattack misclassification and confusion among the three nonzero bias classes. Fig. 6 separates these effects by grouping all predicted attack classes. An attack assigned to an incorrect nonzero class contributes to attack detection but not to exact-severity recall. The corresponding recall and false-alarm statistics are reported in Appendix A-C.

Fig. 7 complements the aggregate confusion-matrix results by showing a representative ACIM detection sequence on a continuous experimental signal. Both sensing locations follow the scheduled normal/FDIA intervals, but the motor-side detector produces cleaner transitions than the PCC-side detector, especially near the first attack onset and low-severity FDIA intervals. These results indicate that motor-side measurements are more separable, while PCC measurements remain useful but less reliable for fine severity discriminationshows a 200-s overview and a 22-s transition detail from a held-out ACIM recording. The first test recording in alphabetical order was used to provide a reproducible example. The predicted scores show both detected attacks and false alarms relative to the recorded injection schedule. The complete recording, including its normal tail, and a corresponding held-out PMSM trace are supplied with the revision materials.

VI. LATENCY AND REDUCED-FEATURE INFERENCE ANALYSIS

Practical PCC monitoring requires low latency and stable online throughputprocessing each measurement window within the acquisition interval. This section reports per-window latency, throughput, pruning, and reduced-feature effects. In particular, the analysis focuses on end-to-end processing latency, online throughput, feature-pruning effects , and practical considerations for deployment at the PCCevaluates processing time and throughput, then examines the effects of feature selection and LF/HF/TD feature-family reduction.

A. Inference Latency and Throughput on CPU/GPU

Runtime was benchmarked on an

Runtime was measured on the Intel Core Ultra 9 285K CPU and NVIDIA RTX 5090 GPU using the selected LF/HF-Inception pipelinesseed-1234 binary models. Each variant used 10 warm-up executions and 50 timed repetitions, with variant order randomized within each repetition. Full, pruned, and ablated inputs used the same selective feature-extraction implementation. The benchmark was split into three stages: (i) LFseparates feature extraction, normalization/HF feature extractionfrom one 2, 000-sample raw window, (ii) model inferenceon the resulting feature tensor, and (iii) end-to-end per-window execution. GPU inference was measured with and without host-to-device transfer.

transfer/inference, and full processing, reporting median and 95th-percentile latency (Table IXsummarizes the measured per-window latencies. For the PCC stream, LF/HF feature extraction required 21.72 ms on CPU and the Inception model needs only 0.71 ms on GPU. When host-to-device transfer was included, the GPU inference stage remained sub-millisecond at 0.74 ms . As a result, the PCC end-to-end latency was only 22.46). The reported processing time excludes the 200-ms acquisition window and is distinct from the delay between attack onset and alarm generation. CPU and GPU-assisted versions use the same selected models.

Full-feature GPU-assisted processing takes 23.03 ms for PCC and 67.81 ms for End-to-End inference. A similar pattern was observed for the motor stream. Feature extraction required 65.25 ms on CPU, while model inference required 0.36 ms on GPU. Including transfer, the GPU inference stage was 0.38 ms, leading to an end-to-end latency of 65.63motor sensing, with 95th percentiles of 25.03 and 76.98 ms. For the tested PCC pipeline, the measured end-to-end latency was 22.46 ms under the reported hardware configuration. In terms of throughput, the end-to-end PCC pipeline sustained 44.53 Both measured processing times are below the 200-ms window interval. Reciprocal median processing rates are approximately 43.42 and 14.75 windows/s, while the corresponding motor pipeline sustained 15.24 windows/srespectively; these are workstation processing capacities, not measured embedded worst-case guarantees. Acquisition, scheduling, and any alarm-persistence logic must be included in a deployed timing budget.

B. Feature Pruning for Low-Latency Anomaly Detection

Feature pruning was evaluated for the fdia2 task to reduce online latency. Features

Binary-model features were ranked by permutation importance, defined as the balanced-accuracy drop caused by permuting each feature column. Reduced subsets were then formed using global top-75%, top-50%, and top-25% rankings, as well as validation-set permutation importance: the decrease in balanced accuracy when one feature trajectory is permuted across validation windows, repeated three times. Test labels were not used for ranking. Within each split, global top-75%, top-50%, and top-25% subsets and per-channel top-15and top-10 selections. Because the benchmark uses selective LF/HF feature computation, only retained features are extracted; therefore, the measured extraction latency reflects the computational benefit of pruningtop-10 subsets were retrained with the full model's selected architecture and training protocol. Fig. 8 summarizes validation importance across repeats; correlated features can share predictive information, so these scores do not identify causal attack mechanisms.

Table X shows that pruning can reduce latency while preserving, and in some cases improving, detection performance. For the PCC stream, the best tradeoff was

ORIGINAL — Figure 5 (main.tex)

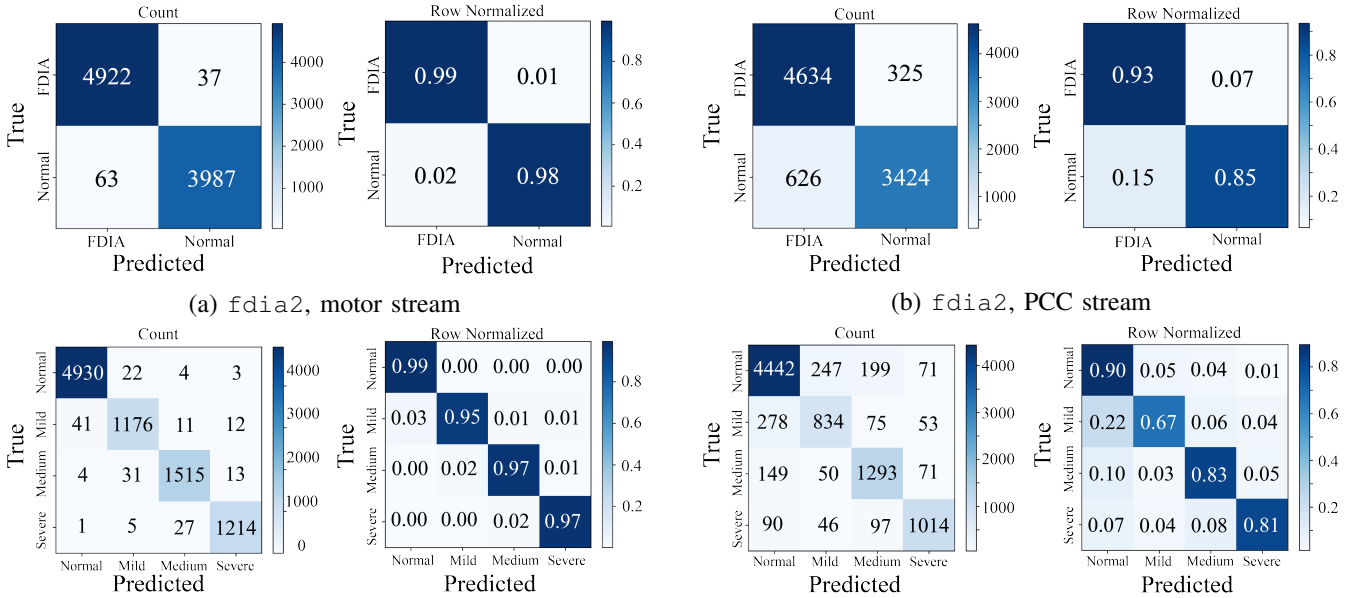


Figure 5: In-domain confusion matrices for the proposed LF/HF InceptionTime classifier: (a) `fdia2` with motor-side measurements, (b) `fdia2` with PCC measurements, (c) `severity4` with motor-side measurements, and (d) `severity4` with PCC measurements.

REVISED — Figure 5 (main_revised.tex)

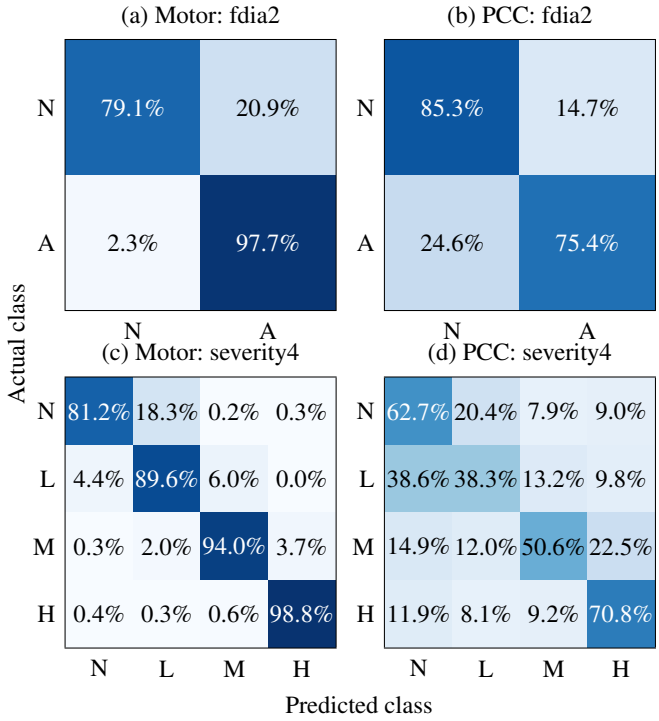


Fig. 5. Confusion matrices for `fdia2` and `severity4`. Rows are actual and columns predicted classes. Percentages normalize pooled prediction counts within each row across three holdouts; repeated appearances of a recording are not independent replications. N/L/M/H denote normal/low/medium/high bias.

ORIGINAL — Table 7 (main.tex)

Table 7: Cross-domain source-only performance without target-domain adaptation on `fdia2` task.

Stream	Direction	Accuracy (%)	Balanced Acc. (%)
Motor	ACIM → PMSM	53.87	50.10
Motor	PMSM → ACIM	51.08	51.80
PCC	ACIM → PMSM	54.79	50.07
PCC	PMSM → ACIM	52.38	57.03

REVISED — Table 7 (main_revised.tex)

TABLE VII

SOURCE-ONLY FDIA2 TRANSFER WITHOUT TARGET ADAPTATION (%).

Direction	Accuracy	BA	Macro F1
MOTOR stream			
ACIM → PMSM	55.21 ± 1.14	50.60 ± 0.73	37.71 ± 2.37
PMSM → ACIM	36.22 ± 4.23	36.08 ± 4.11	33.89 ± 4.62
PCC stream			
ACIM → PMSM	55.88 ± 0.64	51.58 ± 1.52	41.13 ± 5.84
PMSM → ACIM	51.94 ± 6.54	56.00 ± 5.69	43.62 ± 11.38

Mean ± sample SD over three holdouts. Directions are different target-domain tests, so no best direction is highlighted.

ORIGINAL — Table 8 (main.tex)

Table 8: Few-shot domain-adaptation results for `fdia2` using labeled target-domain fine-tuning. Values are mean test balanced accuracy (%) over three seeds.

Stream	Direction	Source-only	1%	5%	10%	20%
Motor	ACIM → PMSM	50.10	50.06	49.63	53.15	57.49
Motor	PMSM → ACIM	51.80	57.95	73.75	78.86	86.10
PCC	ACIM → PMSM	50.07	51.79	55.51	57.47	60.06
PCC	PMSM → ACIM	57.03	70.10	84.56	89.12	91.16

REVISED — Table 8 (main_revised.tex)

TABLE VIII

FEW-SHOT FDIA2 ADAPTATION BY LABELED TARGET-DOMAIN FINE-TUNING: BALANCED ACCURACY (%).

Direction	Source only	1%	5%	10%	20%
MOTOR stream					
ACIM → PMSM	50.60 ± 0.73	50.43 ± 1.74	53.83 ± 1.14	57.51 ± 7.24	**75.99** ± 1.73
PMSM → ACIM	36.08 ± 4.11	64.35 ± 9.23	73.93 ± 8.78	88.04 ± 5.36	**92.63** ± 3.72
PCC stream					
ACIM → PMSM	51.58 ± 1.52	54.01 ± 0.12	54.62 ± 1.76	54.04 ± 1.35	**56.30** ± 3.23
PMSM → ACIM	56.00 ± 5.69	77.43 ± 10.43	84.22 ± 4.62	86.72 ± 4.37	**91.38** ± 2.06

Mean ± sample SD over three holdouts. Bold: highest observed mean within each direction and stream; no test-based configuration selection. Budgets refer to the target episode-group pool.

REVISED — Figure 6 (main_revised.tex)

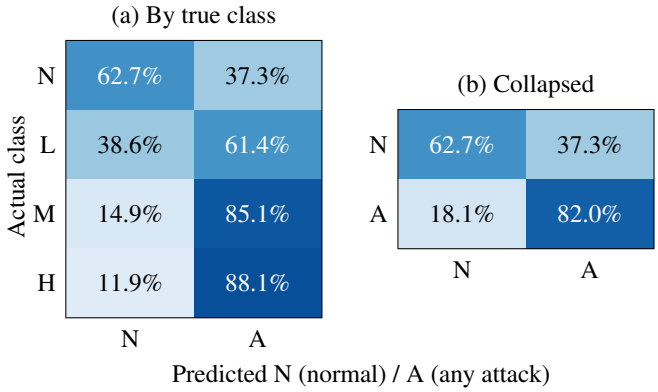
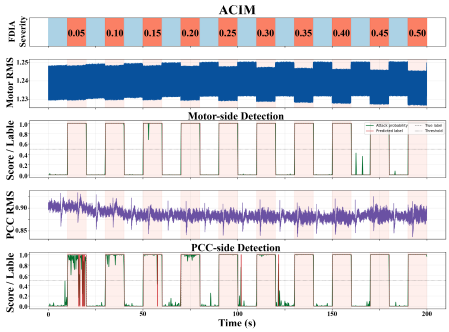
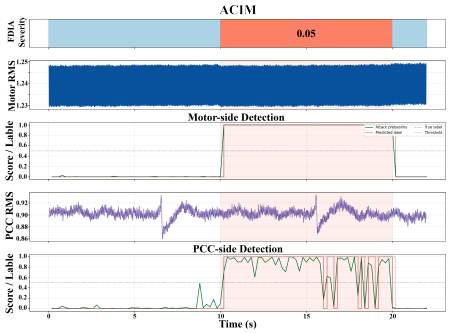


Fig. 6. PCC `severity4` outputs interpreted as (a) normal/any-attack decisions by actual class and (b) a collapsed binary decision. Row percentages pool three holdouts. These decisions come from the severity model, not the separately trained `fdia2` detector.

ORIGINAL — Figure 6 (main.tex)



(a) 200 s overview



(b) 0–22 s zoomed view

Figure 6: Representative continuous-signal FDIA detection example for the ACIM. Panel (a) shows the 200-s sequence, and panel (b) zooms into the minor attack transition.

REVISED — Figure 7 (main_revised.tex)

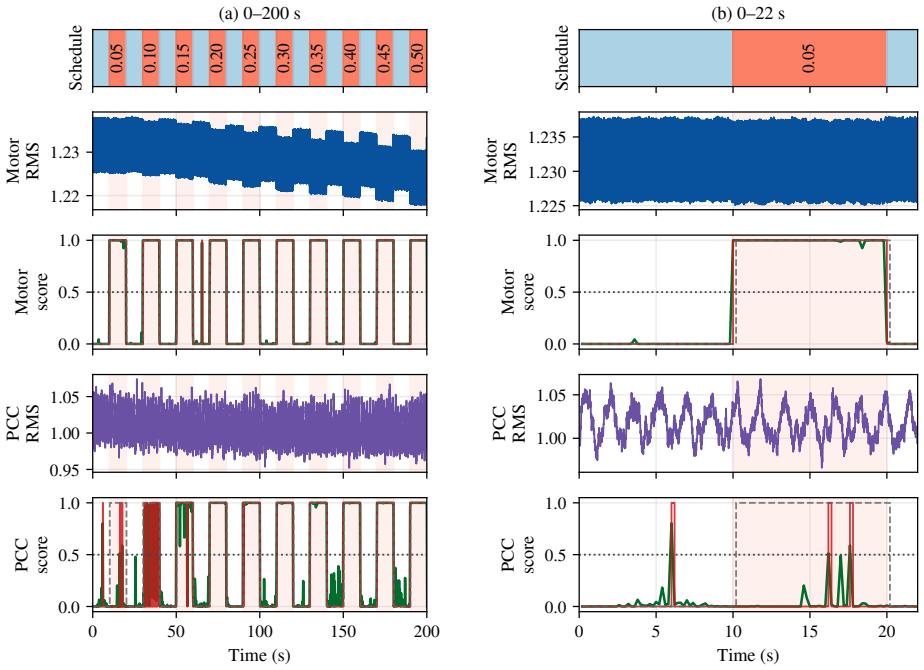


Fig. 7. Held-out ACIM detection: schedule, motor RMS, motor decision, PCC RMS, and PCC decision. (a) First 200 s; (b) first 22 s, including the first low-bias attack. The recording is the first alphabetically among seed-1234 test recordings. Curves use saved held-out predictions at window ends; RMS uses 500-sample frames at stride 25. Green: attack probability; red: predicted label; dashed gray: true label; dotted line: 0.5 threshold. Shading denotes scheduled attacks. Complete-recording and PMSM plots accompany the paper.

obtained with the top-25% subset , which retained quantifies the reduction in processing time obtained by selective feature computation. The PCC top-25% subset retains 18 of 74 features . This subset improved balanced accuracy from 88.99% to 90.11%and reduced end-to-end CPU+GPU latency from 25.38 ms to 19.71 ms , corresponding to a 1.29× speedup. For the motor stream, the top-25% subset retained and approximately preserves balanced accuracy (80.29%), while reducing processing time from 23.03 to 12.44 ms (1.85×). The motor top-25% subset retains 56 of 222 features , improved balanced accuracy from 98.85% and reduces processing time from 67.81 to 99.48% , and reduced latency from 109.07 ms to 56.21 ms, corresponding to a 1.94× speedup.

The intermediate subsets did not always reduce runtime, even when they contained fewer features. This indicates that pruning effectiveness depends not only on featurecount, but also on which LF, HF, and TD feature families remain active. The feature-importance rankings in Fig. 8 further show that the two sensing streams rely on different features: motor-side detection is dominated by TD and selected LF features, whereas PCC-side detection places greater emphasis on harmonic and waveform-shape features. 27.87 ms (2.43×), with balanced accuracy of 86.83% compared with 88.26% for the full input. For each subset, the extractor evaluates only the selected statistics and the intermediate calculations required to obtain them. Shared operations, such as the FFT, are retained when needed by a selected feature; unrelated harmonic, spectral, envelope, and time-domain calculations are omitted. The selected feature values and model outputs were verified against the original implementation, so the trained models and classification results are unchanged. Full and reduced subsets were timed using the same implementation and benchmark conditions. Speedup depends on the computational dependencies of the retained features as well as their number. All evaluated subsets are reported, and the full-feature model remains the reference.

Overall, the LF/HF representation contains redundancy for anomaly detection, and structured pruning can improve both latency and predictive performance. This is especially relevant for PCC-based monitoring, where the compact top-25% subset provides both the fastest and most accurate PCC detector.

C. LF/HF/TD Ablation for Binary Anomaly Detection

A complementary

All seven LF/HF/TD ablation was performed on the `fdia2` taskto identify which feature groups are most useful for binary anomaly detection in the PCC and motor streams. Detailed ablation results are reported in combinations were evaluated by retraining with the selected full-feature architecture fixed within each task, stream, and split. Table XI . For the PCCstream, the reports binary results. For PCC, LF+Time subset achieved 88.67reaches 80.03% balanced accuracy, close to the full model at 88.99%, whereas HF-only and HF+Time dropped to 73.98% and 79.78%, respectively. For the motor stream, the input at 80.29%, while reducing GPU-assisted processing from 23.03 to 12.69 ms (1.81×). The small difference in balanced accuracy suggests that LF and TD features provide most of the useful information for PCC binary detection under the tested conditions.

For motor sensing, time-only subset matched or slightly exceeded the full model, reaching 99.45reaches 92.89% balanced accuracy compared with 98.85and HF+Time reaches 94.11%, compared with 88.26% for the full representation, while input. Time-only processing takes 16.33 ms (4.15× faster than full). HF-only dropped to 65.34% . Overall, the ablation confirms that PCC detection relies mainly on LF+Time information, motor-side detection is driven primarily by TD features, and HF features are the least informative group in both streams. This trend is consistent with the pruning results and supports reduced-feature inference in the tested pipelineperforms poorly for both streams (63.76% motor and 64.49% PCC balanced accuracy), but its combination with time features is useful for motor sensing. These ablations quantify the contribution of each feature family relative to the full-feature reference; they are not used for model selection. Severity ablations and complete per-repeat measurements are supplied with the revision materials.

VII. Discussion

A. Why Adaptation Helps More in One Direction

The adaptation results show that ACIM and PMSM do not form a symmetric transfer pair. PMSM → ACIM adaptation is much more effective than ACIM → PMSM adaptation, particularly for PCC sensing. In the PCC stream, the ACIM → PMSM model tends to collapse toward the normal class, whereas the PMSM → ACIM model retains more attack-sensitive structure and can be corrected with limited labeled ACIM data. Thus, the source motor affects the initial class bias, the retained target-relevant information, and the benefit obtained from few-shot

The PCC results retain a strong direction asymmetry: at 20% target labels, PMSM-to-ACIM fine-tuning . For practical PCC deployment, a small amount of target-domain calibration data is therefore important when the deployed motor family differs from the source training domain. reaches 91.38% balanced accuracy, while ACIM-to-PMSM reaches 56.30%. Source-only predictions also show different class biases: for PCC ACIM-to-PMSM, the mean attack miss rate is 92.71%; in the reverse direction the normal false-alarm rate is 86.03%. These biases show that source-domain decision boundaries and operating distributions do not transfer reliably. The present experiments do not separate the physical and operating-condition contributions to this asymmetry.

The adaptation results indicate that the required target-domain training depends on the transfer direction. In particular, the strong PCC PMSM-to-ACIM gain is not reproduced in the reverse direction. Source-only and fine-tuned results quantify the benefit of updating a source model with target labels; they do not isolate the contribution of source pretraining from the benefit of those labels.

B. PCC vs Motor Tradeoffs

ORIGINAL — Table 9 (main.tex)

Table 9: Measured per-window latency and throughput of the GPU-assisted LF/HF-InceptionTime pipeline.

Stream	Feature Extraction (ms)	GPU Inference (ms)	GPU Transfer + Inference (ms)	End-to-End (ms)	Throughput (windows/s)
PCC	21.72	0.71	0.74	22.46	44.53
Motor	65.25	0.36	0.38	65.63	15.24

REVISED — Table 9 (main_revised.tex)

TABLE IX
GPU-ASSISTED PER-WINDOW LATENCY AND PROCESSING RATE.

Statistic	Feature extract. (ms)	Norm./transfer + inference (ms)	Total processing (ms)	Rate (windows/s)
PCC stream				
Median	22.30	0.78	23.03	43.42
95th pct.	24.15	0.95	25.03	–
MOTOR stream				
Median	66.38	1.42	67.81	14.75
95th pct.	75.52	1.58	76.98	–

Rate is the reciprocal of median total processing time. Stages are timed within each iteration; marginal percentiles need not sum. Acquisition (200 ms) is excluded. CPU measurements are supplied with the results.

ORIGINAL — Table 10 (main.tex)

Table 10: Effect of feature pruning on `fdia2` performance and runtime.

Feature Set	No. Feature	Balanced Accuracy (%)	F1 (%)	Feature Extraction (%)	End-to-End (ms)	Speedup (×)	Runtime Gain(%)
PCC stream							
Full	74	88.99	89.25	24.61	25.38	1.00	0
75%	56	88.65	88.88	22.11	22.91	1.11	11
50%	37	88.90	89.04	22.26	23.00	1.10	10
25%	18	90.11	90.28	18.88	19.71	1.29	29
15/ch.	30	89.33	89.52	33.95	34.70	0.73	-27
10/ch.	20	89.40	89.61	19.09	20.23	1.25	25
Motor stream							
Full	222	98.85	98.88	108.65	109.07	1.00	0
75%	166	99.24	99.27	110.15	110.74	0.98	-2
50%	111	99.39	99.38	112.59	113.20	0.96	-4
25%	56	99.48	99.51	55.84	56.21	1.94	94
15/ch.	90	99.49	99.50	104.02	104.60	1.04	4
10/ch.	60	99.46	99.47	62.19	62.79	1.74	74

REVISED — Table 10 (main_revised.tex)

TABLE X
FEATURE PRUNING FOR `FDIA2`: PERFORMANCE AND RUNTIME.

Subset	No.	BA (%)	F1 (%)	Extract. (ms)	Total (ms)	Speedup (×)
PCC stream						
Full	74	80.29 ± 3.03	80.42 ± 2.96	22.30	23.03	1.00
75%	56	78.28 ± 3.07	78.41 ± 3.37	19.65	20.33	1.13
50%	37	78.87 ± 2.79	79.10 ± 2.97	16.63	17.31	1.33
25%	18	80.29 ± 2.14	80.58 ± 2.29	11.81	**12.44**	1.85
15/ch.	30	77.59 ± 4.75	77.74 ± 4.94	15.64	16.25	1.42
10/ch.	20	**80.47** ± 2.79	**80.69** ± 2.80	12.07	12.68	1.82
MOTOR stream						
Full	222	88.26 ± 9.99	87.23 ± 11.39	66.38	67.81	1.00
75%	166	87.95 ± 11.38	87.19 ± 12.42	56.28	57.70	1.18
50%	111	89.28 ± 11.33	88.20 ± 12.87	45.50	46.89	1.45
25%	56	86.83 ± 10.93	85.44 ± 12.39	26.57	**27.87**	2.43
15/ch.	90	**92.10** ± 9.71	**91.44** ± 10.88	37.68	38.96	1.74
10/ch.	60	88.18 ± 9.28	87.20 ± 10.62	28.48	29.80	2.28

Performance: mean $\pm$ sample SD. Timing: seed-1234 GPU-assisted median; 95th percentiles are supplied in the results. Bold: highest BA/F1 or lowest total latency within a stream. Only selected statistics and their required intermediate calculations are evaluated.

ORIGINAL — Table 11 (main.tex)

Table 11: LF/HF/TD ablation results for `fdia2`.

Feature Group	Selected Features	Accuracy (%)	Balanced Accuracy (%)	F1 (%)
PCC stream				
Full	74	89.44	88.99	89.25
LF only	36	88.31	87.89	88.11
HF only	22	74.43	73.98	74.06
Time only	16	80.48	79.71	79.99
LF + HF	58	84.04	83.66	83.80
LF + Time	52	89.22	88.67	88.99
HF + Time	38	80.19	79.78	79.89
Motor stream				
Full	222	98.89	98.85	98.88
LF only	108	89.77	89.48	89.62
HF only	66	66.06	65.34	65.41
Time only	48	99.47	99.45	99.46
LF + HF	174	88.98	88.63	88.81
LF + Time	156	99.03	99.00	99.02
HF + Time	114	99.07	99.00	99.06

REVISED — Table 11 (main_revised.tex)

TABLE XI
LF/HF/TD ABLATION FOR `FDIA2` (%).

Features	No.	Accuracy	BA	Macro F1
PCC stream				
Full	74	**80.78** ± 2.95	**80.29** ± 3.03	**80.42** ± 2.96
LF only	36	74.03 ± 3.84	73.56 ± 3.13	73.59 ± 3.33
HF only	22	64.50 ± 4.62	64.49 ± 4.56	64.21 ± 4.71
Time only	16	72.18 ± 1.73	70.66 ± 1.77	70.65 ± 1.76
LF + HF	58	68.66 ± 4.60	68.15 ± 4.26	68.18 ± 4.39
LF + Time	52	80.49 ± 1.73	80.03 ± 1.79	80.16 ± 1.73
HF + Time	38	69.49 ± 3.66	67.85 ± 3.11	67.72 ± 3.06
MOTOR stream				
Full	222	87.41 ± 11.13	88.26 ± 9.99	87.23 ± 11.39
LF only	108	81.61 ± 3.08	80.54 ± 2.55	80.94 ± 2.77
HF only	66	63.32 ± 4.96	63.76 ± 4.28	63.14 ± 4.81
Time only	48	92.61 ± 5.51	92.89 ± 4.96	92.58 ± 5.50
LF + HF	174	80.33 ± 7.88	79.91 ± 7.06	80.05 ± 7.60
LF + Time	156	90.70 ± 8.32	91.17 ± 7.45	90.65 ± 8.35
HF + Time	114	**93.87** ± 4.56	**94.11** ± 4.04	**93.85** ± 4.54

Mean $\pm$ sample SD over three holdouts. Bold: highest mean within each stream; these test comparisons do not select a replacement primary model.

The results reveal a consistent tradeoff between sensing convenience and predictive separability. Motor-side measurements provide the clearest attack-sensitive signatures across nearly all tasks. In the in-domain experiments, the motor stream outperformed the PCC stream for both `fdia2` and `severity4`, and the confusion matrices showed substantially stronger diagonal structure for motor-side classification. The motor stream also achieved lower regression error in the attack-only severity-regression task, indicating that direct motor-current sensing preserves richer information about attack intensity and progression.

PCC remains the more deployment-oriented option because it avoids direct terminal access and matches shared industrial power infrastructure. Despite lower separability, PCC achieved useful in-domain detection and strong adapted performance, especially for PMSM → ACIM transfer.

Overall, the motor stream Motor sensing provides higher mean class-balanced performance on this dataset, particularly for distinguishing injected-bias bins. PCC sensing reduces terminal instrumentation but aggregates the responses of both drive branches and their shared converter interface. For the four-class PCC model, relatively high any-attack recall at medium/high bias coexists with low exact-severity recall and frequent false alarms. The separate binary detector is the stronger sensing modality when maximum classification performance is the priority, especially for fine-grained severity discrimination. The PCC stream, by contrast, is the more deployment-oriented choice because it avoids invasive sensing while still supporting accurate and computationally efficient FDIA detection. The value of the framework is therefore not in surpassing motor-side sensing, but in improving the accuracy and computational efficiencyof PCC-based monitoring on the tested platformprimary PCC alarm model in these experiments.

The motor-type task is an auxiliary diagnostic of distinguishable recording domains, not proof of a causal motor-physics explanation: the motor families also have different load-setting ranges and share only one installation per family. Attack-only regression likewise estimates injected bias rather than torque or damage (Appendix A-E). For efficiency, the PCC LF+Time representation offers a measured alternative to full features with similar balanced accuracy. Neither classification scores nor processing latency alone establish safe automatic mitigation.

C. Limitations

Several limitations should be noted when interpreting the present results. First, the study considers only two motor families, ACIM and PMSM, under a fixed experimental FDIA framework. Broader validation across additional machines, loads, control modes, sensing setups, and attack families is needed before generalizing beyond the tested platform. The evidence is limited to additive, piecewise-constant current-feedback bias on one two-branch laboratory platform. Monitoring sensors are assumed uncompromised; coordinated attacks on those sensors, replay, denial of service, command attacks, and stealthy time-varying perturbations were not tested. Three fixed speed settings and three load settings were recorded for each machine, but dedicated speed ramps, load steps, and other dynamic operating conditions were not evaluated. Brake excitation identifies operating settings; calibrated shaft torque and mechanical harm were not measured.

Second, the domain-adaptation analysis is based on supervised few-shot fine-tuning with labeled target-domain data . This does not cover cases where target labels are unavailable or delayed. Unsupervised or self-supervised adaptation strategies may therefore be required for more constrained deployment scenarios. PCC signatures can depend on the shared rectifier, downstream inverter topology, control behavior, modulation, sensor transfer functions, and supply impedance. These factors were not varied independently in the dataset. Consequently, no robustness claim is made for other converter topologies, modulation strategies, grids, or installations. Whole-recording holdouts remove within-repeat recording/episode overlap, but remain a retrospective evaluation on reused data from the same apparatus. Three repeated holdouts do not provide independent hardware replication or a strong significance test.

Third, the deployment analysis was performed using a GPU-assisted software implementation and focused mainly on the binary FDIA detection task; embedded implementation and task-specific optimization for severity assessment remain future work.

Finally, some baselines were competitive under macro-F1; therefore, the conclusions are strongest for accuracy and balanced accuracy, not universal model dominanceThe adaptation study requires target labels and uses fixed training budgets; unsupervised adaptation is outside its scope. PCC severity classification has limited balanced accuracy, and its false-alarm rate remains substantial despite higher detection rates for medium- and high-bias attacks. Finally, measured workstation processing times exclude acquisition and do not establish embedded worst-case latency or a tested closed-loop mitigation system.

VIII. Conclusion

This paper experimentally evaluated PCC-based detection of low-severity current-feedback FDIAs in a

This paper presented a nonintrusive FDIA monitoring framework for heterogeneous ACIM/PMSM inverter-fed drive platformdrives. Motor-terminal measurements provided the strongest separability, but PCC measurements supported useful nonintrusive detection despite aggregation, harmonics, and grid-side interactions. In the mixed-domain setting, the proposed LF/HF-InceptionTime method achieved strong `fdia2` detection from both sensinglocations, while PCC `severity4` estimates were less reliable and should be interpreted as coarse diagnostics. Cross-motor experiments showed substantial ACIM/PMSM domain shift, making source-only transfer unreliable. Few-shot target-domain fine-tuning improved transfer performance, with much larger gains for PMSM → ACIM adaptation than for ACIM → PMSM adaptation. Runtime and pruning results further

~~indicate that the PCC pipeline is compatible with the required online processing rate. Overall, the results indicate that PCC-based FDIA monitoring is promising~~ and PCC measurements were evaluated independently using three training–validation–test partitions with disjoint recordings and episodes. Mean binary detection accuracy was 87.41% for motor-terminal sensing and 80.78% for PCC sensing, while four-class severity balanced accuracy was 90.81% and 55.62%, respectively. The PCC results show that detecting an attack is less demanding than assigning its exact bias class; both measures must be considered together with the false-alarm rate. Direct cross-motor transfer was limited, and the improvement obtained by supervised adaptation depended on the transfer direction and sensing location. For PCC detection, LF and time-domain features retained approximately the full-feature balanced accuracy while reducing processing time from 23.03 to 12.69 ms on the tested ~~ACIM/PMSM platform under~~ workstation. These findings identify the practical tradeoffs among sensing location, severity resolution, target-domain adaptation, and processing time for additive current-feedback ~~attacks~~bias monitoring. Validation on additional installations and dynamic operating conditions remains necessary to establish broader applicability.

References

[1] M. Ghaderi, K. Gheitasi, and W. Lucia, "A blended active detection strategy for false data injection attacks in cyber-physical systems," *IEEE Transactions on Control of Network Systems*, vol. 8, no. 1, pp. 168–176, 2021.

[2] B. Yang, J. Ye, and L. Guo, "Fast detection for cyber threats in electric vehicle traction motor drives," *IEEE Transactions on Transportation Electrification*, vol. 8, no. 1, pp. 767–777, 2022.

[3] B. Yang, S. Wu, K. Hu, J. Ye, W. Song, P. Ma, J. Shi, and P. Liu, "Enhanced cyber-attack detection in intelligent motor drives: A transfer learning approach with convolutional neural networks," *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, vol. 5, no. 2, pp. 710–719, 2024.

[4] Y. Yu, Y. Zhao, B. Wang, X. Huang, and D. Xu, "Current sensor fault diagnosis and tolerant control for vsi-based induction motor drives," *IEEE Transactions on Power Electronics*, vol. 33, no. 5, pp. 4238–4248, 2018.

[5] B. Yang, L. Guo, F. Li, J. Ye, and W. Song, "Vulnerability assessments of electric drive systems due to sensor data integrity attacks," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 5, pp. 3301–3310, 2020.

[6] D. Issicaba, A. Simões Costa, and J. L. Colombo, "Real-time monitoring of points of common coupling in distribution systems through state estimation and geometric tests," *IEEE Transactions on Smart Grid*, vol. 7, no. 1, pp. 9–18, 2016.

[7] T. C. Krause, Ł. Huchel, D. H. Green, K. Lee, and S. B. Leeb, "Nonintrusive motor current signature analysis," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–13, 2023.

[8] G. Liang, J. Zhao, F. Luo, S. R. Weller, and Z. Y. Dong, "A review of false data injection attacks against modern power systems," *IEEE Transactions on Smart Grid*, vol. 8, no. 4, pp. 1630–1638, 2017.

[9] A. Basati, J. M. Guerrero, J. C. Vasquez, N. Bazmohammadi, and S. Golestan, "A data-driven framework for fdi attack detection and mitigation in dc microgrids," *Energies*, vol. 15, no. 22, 2022.

[10] S. Nath, I. Akingeneye, J. Wu, and Z. Han, "Quickest detection of false data injection attacks in smart grid with dynamic models," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 10, no. 1, pp. 1292–1302, 2022.

[11] O. A. Beg, T. T. Johnson, and A. Davoudi, "Detection of false-data injection attacks in cyber-physical dc microgrids," *IEEE Transactions on Industrial Informatics*, vol. 13, no. 5, pp. 2693–2703, 2017.

[12] A. Sargolzaei, K. Yazdani, A. Abbaspour, C. D. Crane III, and W. E. Dixon, "Detection and mitigation of false data injection attacks in networked control systems," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 6, pp. 4281–4292, 2020.

[13] A. Takiddin, R. Atat, M. Ismail, O. Boyaci, K. R. Davis, and E. Serpedin, "Generalized graph neural network-based detection of false data injection attacks in smart grids," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 7, no. 3, pp. 618–630, 2023.

[14] M. Shinde-Pawar, D. Shahane, K. Kadam, M. Sharma, M. Yaramadhi, and R. Maranan, *Machine learning-driven architecture for cyber-attack detection and motor status monitoring for securing industrial IoT*, pp. 702–707. 01 2025.

[15] H. I. Fawaz, B. Lucas, G. Forestier, C. Pelletier, D. F. Schmidt, J. Weber, G. I. Webb, L. Idoumghar, P.-A. Muller, and F. Petitjean, "Inceptiontime: Finding alexnet for time series classification," *Data Mining and Knowledge Discovery*, vol. 34, no. 6, pp. 1936–1962, 2020.

[16] C. W. Tan, A. Dempster, C. Bergmeir, and G. I. Webb, "MultiRocket: Multiple pooling operators and transformations for fast and effective time series classification," *Data Mining and Knowledge Discovery*, vol. 36, no. 5, pp. 1623–1646, 2022.

Appendix A
Additional Experimental and Diagnostic Results

This appendix provides ~~supplementary material supporting the main results, including representative FDIA episodes and the extracted LF/HF/TD feature set~~representative FDIA recordings, the extracted feature definitions, severity-conditioned detection metrics, hyperparameter sensitivity, and auxiliary bias regression. Complete numeric results and per-repeat configurations accompany the manuscript.

A. Representative FDIA Episodes

Figure 9 shows representative consecutive FDIA episodes applied to the ACIM and PMSM drive systems. In both examples, the top panel shows the scheduled FDIA severity level, while the middle and bottom panels show the corresponding motor-current and PCC-current RMS responses during the first 200 ~~s of operation~~s of each selected recording. RMS uses 500-sample frames at stride 25, matching the time-domain RMS definition. Complete-recording diagnostics retain the post-schedule normal tails.

B. Extracted Feature Set

Table XII summarizes the extracted features used in the proposed LF/HF/TD pipeline. Each channel is represented using low-frequency spectral features, high-frequency spectral-envelope features, and time-domain statistical features.

C. Severity-Conditioned Attack Detection

Table XIII separates correct bias-bin classification from detecting any attack. The repeated false-alarm entries belong to the same four-class model and do not depend on the actual attack level. Fig. 6 presents the corresponding PCC decision matrices; pooled normal recall can differ slightly from the mean per-repeat recall because normal-window counts vary across holdouts.

D. Hyperparameter Sensitivity

Fig. 10 reports all one-factor perturbations. PCC binary mean balanced accuracy ranges from 72.23% to 78.33% among these variants, compared with 80.29% for the selected reference. PCC severity varies from 51.25% to 55.71%,

ORIGINAL — Figure 7 (main.tex)

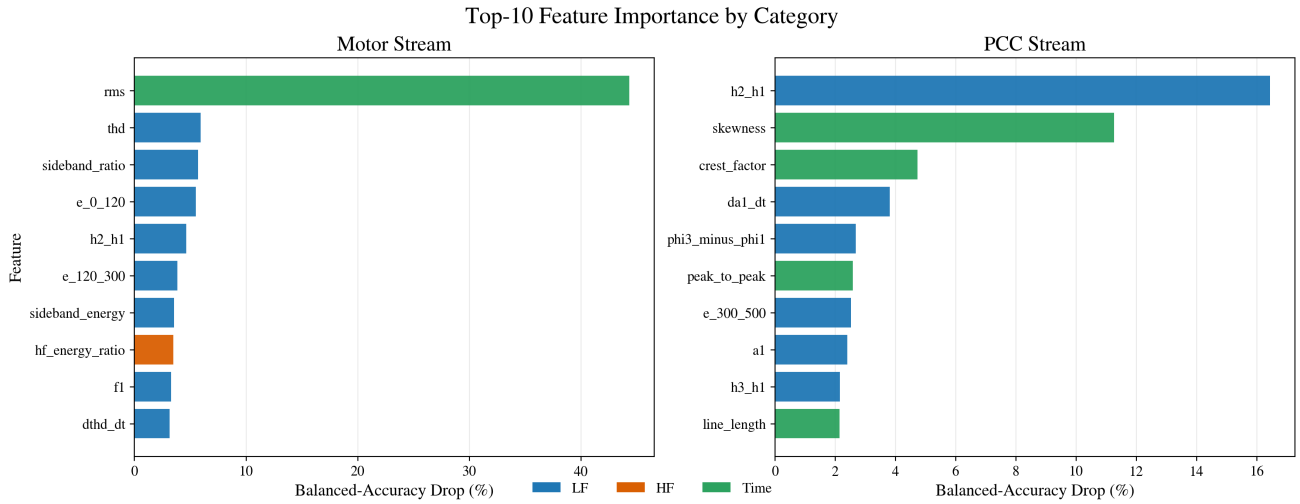


Figure 7: Feature-importance rankings for the motor and PCC streams.

REVISED — Figure 8 (main_revised.tex)

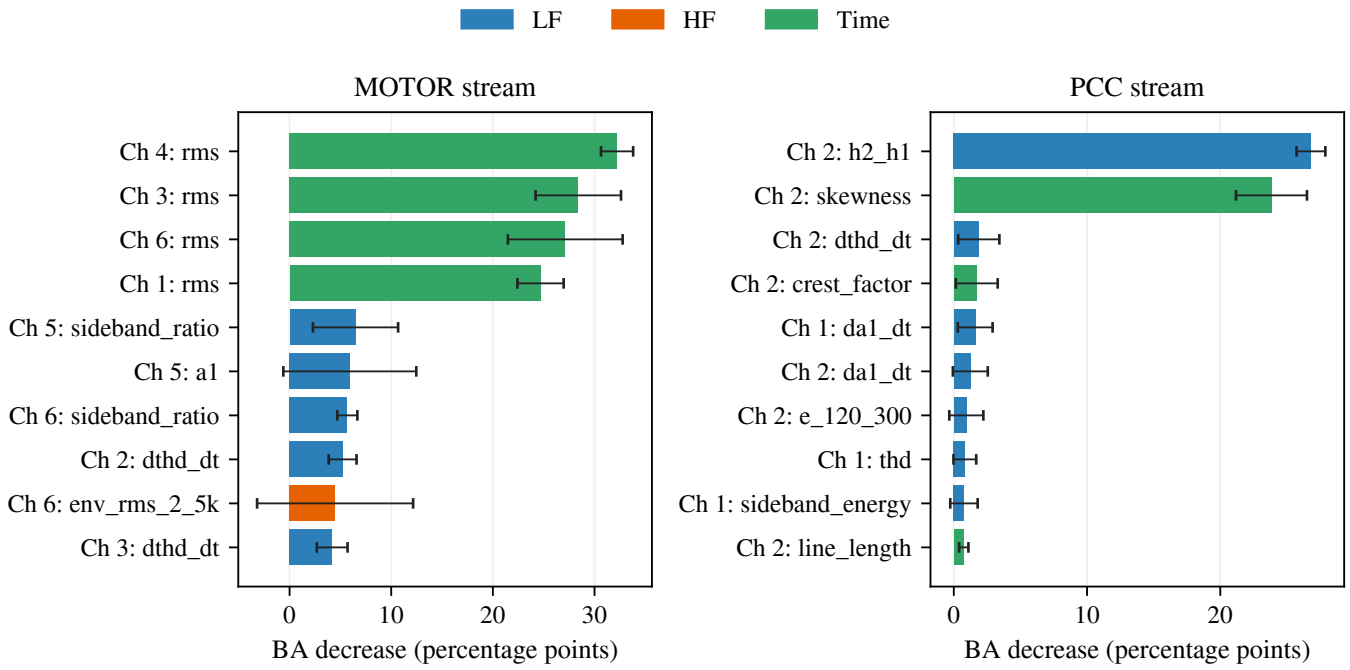


Fig. 8. Top ten channel-specific features ranked by mean validation permutation importance. Blue, orange, and green bars denote LF, HF, and time-domain features, respectively. Bar length is the mean balanced-accuracy decrease; black whiskers indicate one SD across three holdouts. Channels 1–3/4–6 are ACIM/PMSM currents; PCC channels 1/2 are voltage/current.

ORIGINAL — Table 12 (main.tex)

Table 12: Extracted features per channel in the proposed LF/HF/TD pipeline.

Type	No.	Window / Stride	Features
LF	18	500 / 25	A_1, f_1, ϕ_1, THD, H_2/H_1, H_3/H_1, H_5/H_1, H_7/H_1, $\phi_3 - \phi_1$, $\phi_5 - \phi_1$, sideband energy/ratio, E_{0-120}, $E_{120-300}$, $E_{300-500}$, df_1/dt, dA_1/dt, $d\mathrm{THD}/dt$
HF	11	500 / 25	RMS_{2-5k}, energy ratio, centroid, bandwidth, flatness, kurtosis, peak frequency, peak-to-band, envelope RMS, envelope kurtosis, envelope crest factor
TD	8	500 / 25	RMS, standard deviation, crest factor, kurtosis, skewness, peak-to-peak, line length, Teager MAV
Total	37	–	LF (18), HF (11), TD (8)

REVISED — Table 12 (main_revised.tex)

TABLE XII
EXTRACTED FEATURES PER CHANNEL IN THE PROPOSED LF/HF/TD PIPELINE.

Type	No.	Window / Stride	Features
LF	18	500 / 25	A_1, f_1, ϕ_1, THD, H_2/H_1, H_3/H_1, H_5/H_1, H_7/H_1, $\phi_3 - \phi_1$, $\phi_5 - \phi_1$, sideband energy/ratio, E_{0-120}, $E_{120-300}$, $E_{300-500}$, df_1/dt, dA_1/dt, $d\mathrm{THD}/dt$
HF	11	250 / 29	RMS_{2-5k}, energy ratio, centroid, bandwidth, flatness, kurtosis, peak frequency, peak-to-band, envelope RMS, envelope kurtosis, envelope crest factor
TD	8	500 / 25	RMS, standard deviation, crest factor, kurtosis, skewness, peak-to-peak, line length, Teager MAV
Total	37	–	LF (18), HF (11), TD (8)

ORIGINAL — Figure 8 (main.tex)

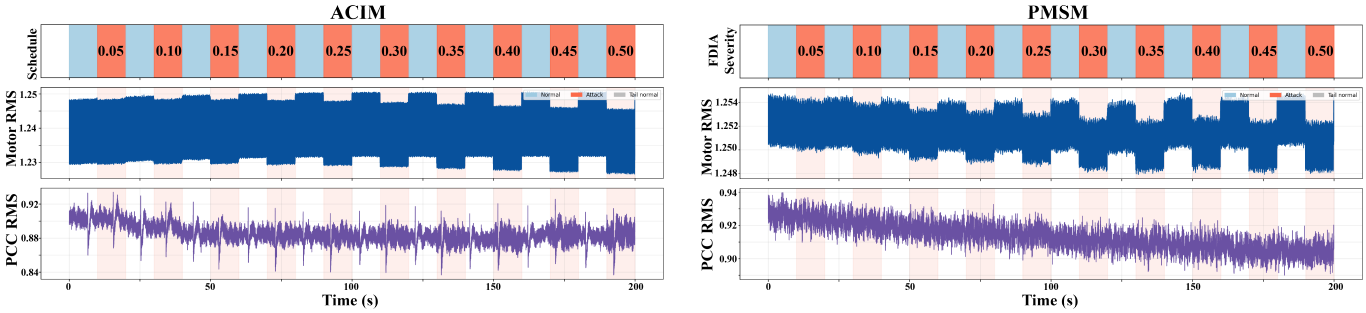


Figure 8: Representative consecutive FDIA episodes applied to the ACIM and PMSM motor-drive systems. Top: scheduled FDIA severity; middle: motor-current RMS; bottom: PCC-current RMS.

REVISED — Figure 9 (main_revised.tex)

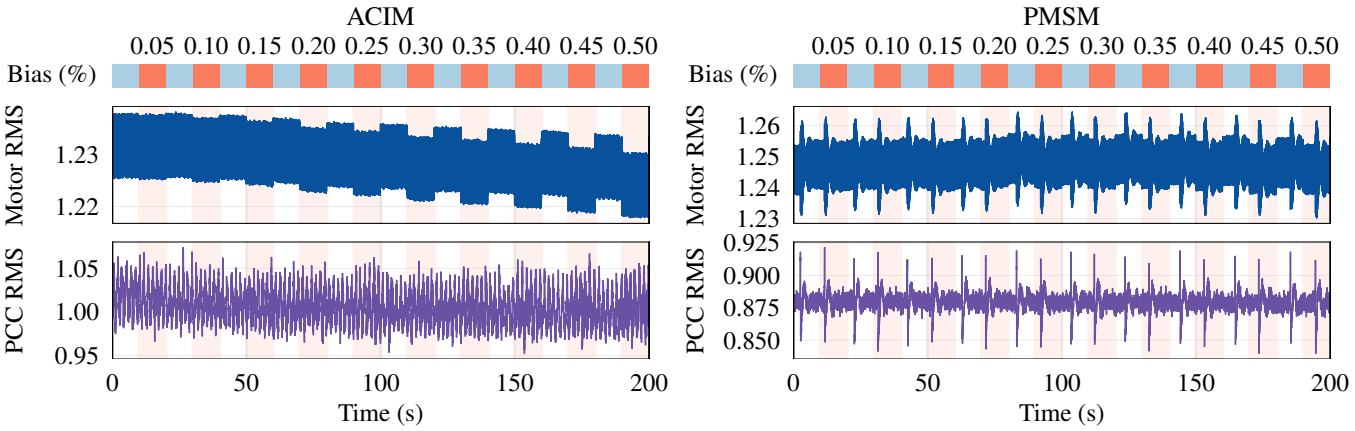


Fig. 9. First 200 s of prespecified held-out ACIM and PMSM recordings, showing the schedule and current RMS. Blue and red schedule blocks denote normal and attack intervals; numbers give maximum injected bias (%). Middle/bottom: motor/PCC RMS over 500-sample frames at stride 25, in acquisition units. These are descriptive electrical traces, not calibrated torque measurements.

REVISED — Table 13 (main_revised.tex)

TABLE XIII
EXACT-SEVERITY AND ANY-ATTACK RECALL FROM FOUR-CLASS PREDICTIONS (%).

True bias	Exact recall	Any-attack recall	Normal false alarms
MOTOR stream			
Low	89.58 ± 2.14	95.58 ± 4.15	19.08 ± 16.08
Medium	93.96 ± 1.05	99.71 ± 0.04	19.08 ± 16.08
High	98.76 ± 0.27	99.65 ± 0.29	19.08 ± 16.08
PCC stream			
Low	38.33 ± 7.04	61.39 ± 2.64	37.23 ± 3.17
Medium	50.58 ± 7.01	85.08 ± 6.52	37.23 ± 3.17
High	70.79 ± 2.28	88.09 ± 7.22	37.23 ± 3.17

Mean $\pm$ sample SD over three holdouts. False alarms refer to the same normal class for all three attack levels. Different metrics are not competing methods, so none is highlighted as best.

compared with 55.62%. Motor severity is more sensitive to the high learning rate, falling to 77.69% mean balanced accuracy. The PCC severity limitation persists across the evaluated settings, although classification performance remains sensitive to the training parameters.

E. Auxiliary Diagnostic Tasks

The auxiliary motor-type classifier (`motor2`) distinguishes ACIM and PMSM recording conditions using a two-class softmax output. Its results are included in Table V. For attack-only regression, the target is $100\|\boldsymbol{\alpha}\|_\infty$, in percentage points of nominal current-feedback bias. A scalar output replaces the classification head; training uses Smooth-L1 loss with transition parameter 0.5 and validation mean absolute error (MAE) for checkpoint selection. The severity-selected architecture is retained within each stream and repeat. Motor-side MAE is 0.0196 ± 0.0041 percentage points, with mean $R^2 = 0.961$; PCC MAE is 0.0981 ± 0.0049 percentage points, with mean $R^2 = 0.267$. Fig. 11 shows prediction distributions at every tested bias level for each repeat. These are estimates of injected amplitude on attacked windows, not measured mechanical impact.

REVISED — Figure 10 (main_revised.tex)

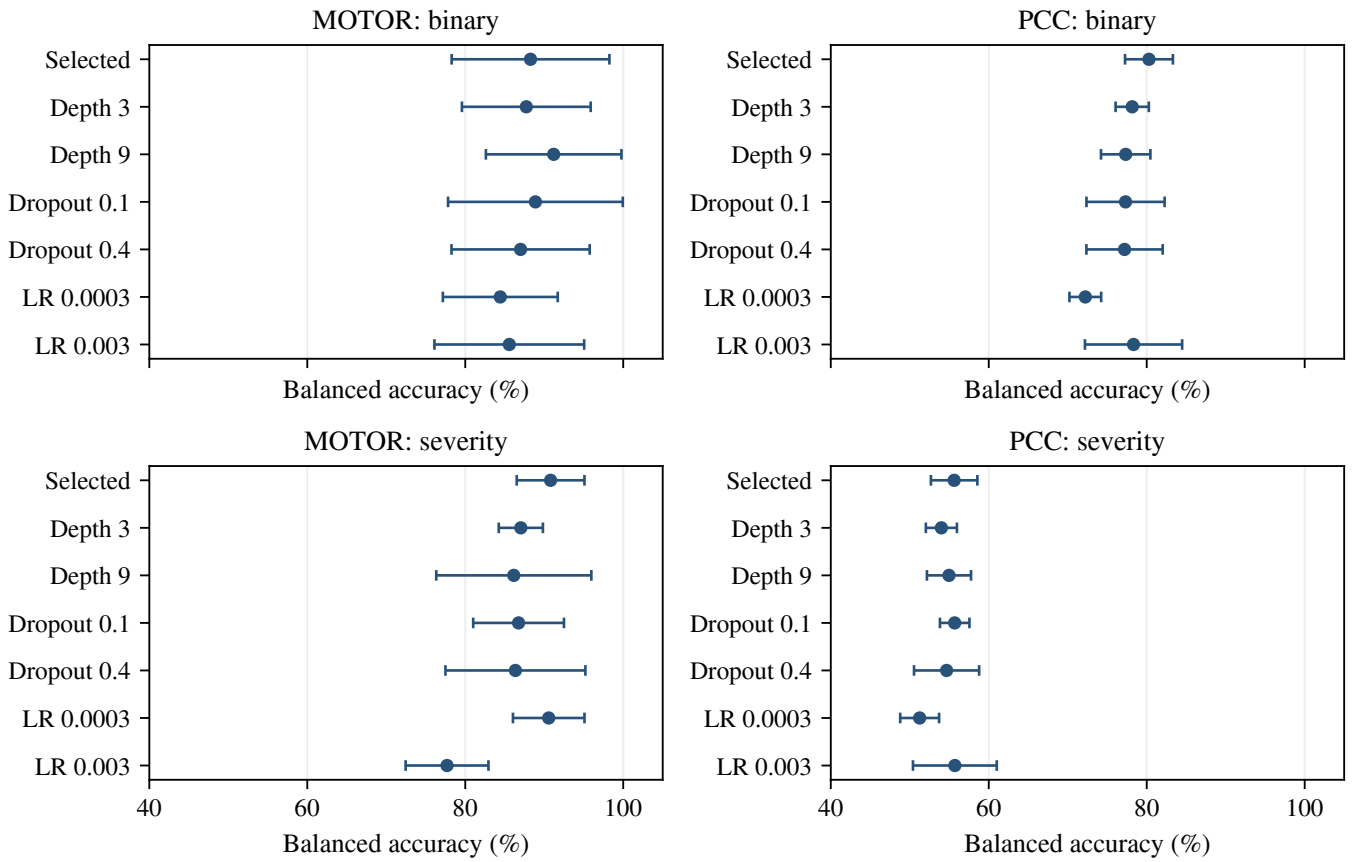


Fig. 10. One-factor sensitivity around the validation-selected full-feature configuration in each repeat. Error bars indicate sample SD across three recording holdouts. Only the named hyperparameter changes; the selected reference can differ across tasks, streams, and repeats.

REVISED — Figure 11 (main_revised.tex)

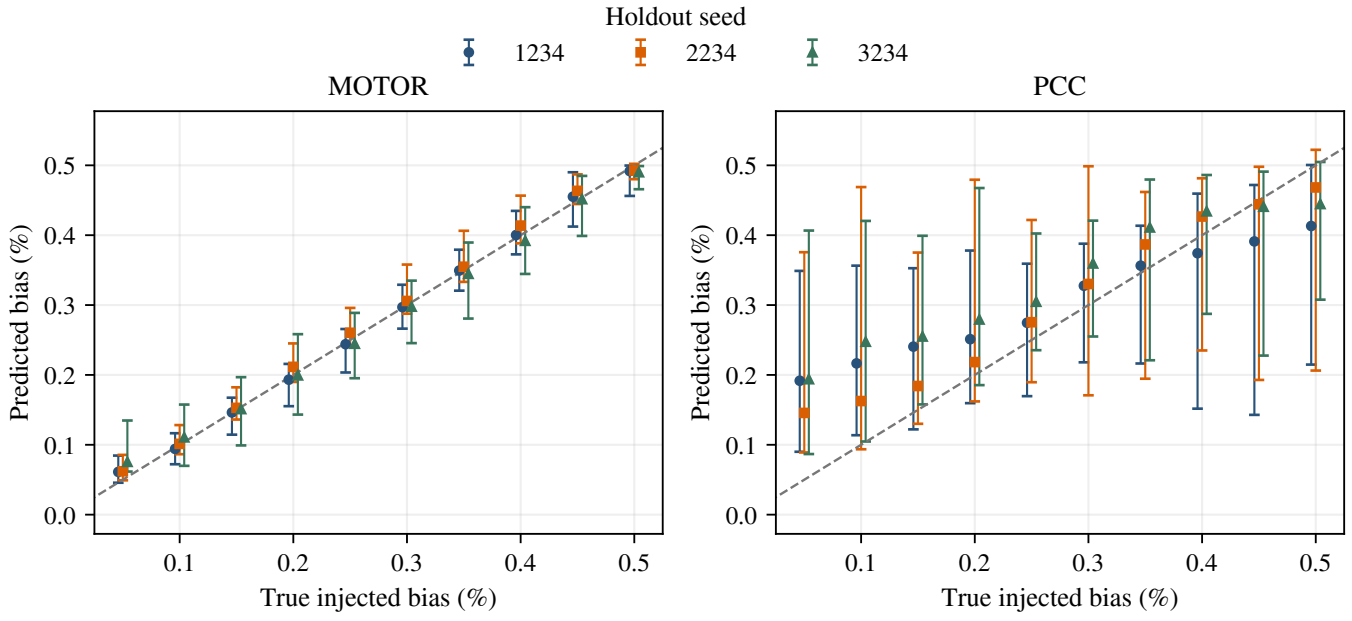


Fig. 11. Attack-only injected-bias regression on held-out recordings. Points and whiskers show the median and 10th–90th percentiles of predictions at each true bias, separately for each holdout seed; slight horizontal offsets separate repeats. The dashed diagonal is ideal agreement. All held-out attack windows are included; repeated appearances across holdouts are not independent recordings.